

APPROVED: 03 October 2024  
doi: 10.2903/sp.efsa.2024.EN-9062

# Alternative protein sources in the European diets – integrating health risk-benefit and sustainability

## ‘ALTERNATIVA’

Ricardo Assunção<sup>1</sup>; Beatrice Biasini<sup>2</sup>; Géraldine Boué<sup>3</sup>; Ramona Clep<sup>4</sup>; Inês Coelho<sup>1</sup>; Elena Cozzi<sup>2</sup>; Adriana Deacu<sup>4</sup>; Lea Jakobsen<sup>5</sup>; Carla Martins<sup>1</sup>; Carla Motta<sup>1</sup>; Androniki Naska<sup>6</sup>; Aikaterini Niforou<sup>6</sup>; Marta Pavel<sup>4</sup>; Sara Pires<sup>5</sup>; Morten Poulsen<sup>5</sup>; Ana Serôdio<sup>1</sup>; Dana Sparatu<sup>4</sup>; Alexandru Supeanu<sup>4</sup>; Susana Santiago<sup>1</sup>; Marta Ventura<sup>1</sup>; Davide Menozzi<sup>2</sup>

<sup>1</sup>Food and Nutrition Department, National Institute of Health Doutor Ricardo Jorge, Lisbon, Portugal; <sup>2</sup>University of Parma, Parma, Italy; <sup>3</sup>French National Institute for Agriculture, Food and Environment, Nantes, France; <sup>4</sup>National Sanitary Veterinary and Food Safety Authority, Romania\*; <sup>5</sup>National Food Institute, Technical University of Denmark, Lyngby, Denmark; <sup>6</sup>National and Kapodistrian University of Athens, Athens, Greece

\* This partner participated only in the development of the project proposal and in the first stages of the project implementation

## Abstract

The pressure global food systems exert on the environment has amplified over time, following the growth of the human population, changes in the food consumption patterns and intensification of the production systems. Without changes in food systems, an overpressure on agriculture and food production will be unavoidable, with inevitable consequences for the environment and human health. Animal-derived protein production like meat and milk is among the most important drivers of this impact. Under this context, the evaluation of the health and sustainability impacts of alternatives for animal-based food proteins are central aspects of this discussion. Health risk-benefit assessment (RBA) of foods estimates the overall human health impact following exposure to a particular food or food component, integrating risks and benefits in comparable measures. Methods that simultaneously consider the environmental impact of the food production systems were needed to perform a sustainable holistic assessment. “ALTERNATIVA | Alternative protein sources in the European diets – integrating health risk-

benefit and sustainability” aimed to develop a holistic approach to assess the impact of alternative protein sources, integrating health and sustainability assessment. This project, considering five complementary work packages, developed activities to enhance the capacity-building among members of the consortium in applying methodologies related to the health RBA and sustainability impact assessment; to develop, in collaboration with different stakeholders, a protocol to integrate the assessment of the impact of foods/diets on health and on sustainability; to apply the derived framework to a case study, to validate it with a real case scenario, estimating the health and sustainability impacts of replacing red meat consumption by pulses in two different countries; and to disseminate and promote the established method of integrated health and sustainability assessment to other potential EU users. This external scientific report summarizes the main outputs and results obtained in the context of the ALTERNATIVA project.

© Copyright with beneficiaries, 2024

**Key words:** Food systems; Alternative protein sources; Health; Economic and Social; RBA Environment; Sustainability.

**Question number:** EFSA-Q-2024-00602

**Correspondence:** [ricardo.assuncao@insa.min-saude.pt](mailto:ricardo.assuncao@insa.min-saude.pt)

**Disclaimer:** The present document has been produced and adopted by the bodies identified above as author(s). In accordance with Article 36 of Regulation (EC) No 178/2002, this task has been carried out exclusively by the author(s) in the context of a grant agreement between the European Food Safety Authority and the author(s). The present document is published complying with the transparency principle to which the Authority is subject. It cannot be considered as an output adopted by the Authority. The European Food Safety Authority reserves its rights, view and position as regards the issues addressed and the conclusions reached in the present document, without prejudice to the rights of the authors.

**Acknowledgements:** The project Consortium would like to acknowledge the contribution of João Pedro Melo and Prof. Ana Vieira from the Instituto Superior Técnico, University of Lisbon, for all their support in the implementation of the case study, especially in the application of the Multi-Criteria Decision Analysis methodology. The project Consortium is also grateful to Prof. Penny Panagiotopoulou of the University of Patras for assistance in assessing the stakeholders' current understanding and attitudes towards a health risk-benefit assessment and sustainability aspects. The project Consortium would also like to acknowledge the support from EFSA staff, namely Martina Kurišová and Ruth Roldán Torres.

**Suggested citation:** Assunção R., Biasini B., Boué G., Coelho I., Cozzi E., Jakobsen L., Martins C., Motta C., Naska A., Niforou A., Pires S., Poulsen M., Serôdio A., Santiago S., Ventura M., Menozzi D., 2024. Alternative protein sources in the European diets – integrating health risk-benefit and sustainability - ALTERNATIVA. EFSA supporting publication 2024:EN-9062. 59 pp. doi:10.2903/sp.efsa.2024.EN-9062

**ISSN:** 2397-8325

© Copyright with beneficiaries, 2024

Reproduction of the images listed below is prohibited and permission must be sought directly from the copyright holder.

## Summary

The pressure global food systems exert to the environment has amplified over time, following growth of the human population, changes in the food consumption patterns and the intensification of the production systems. By 2050, a growth of more than 30% of the current global population is expected. This increase will be followed by an increase in demand for food. Although technological innovation contributed to increase food availability, security, and safety with a consequential reduction in hunger, poverty and mortality worldwide, the agri-food system pressure on the environment has amplified over time. Therefore, without virtuous transformations, an overpressure on agriculture and food production is unavoidable, with detrimental consequences for the environment and human health. Food production, primarily agriculture, causes negative externalities, such as the contribution to climate change, exploitation and pollution of land and water bodies, which are among the largest drivers of biodiversity loss. In particular, animal-derived protein production like ruminant meat and dairy contributes at most to the environmental impact. From the consumption side, unhealthy diets are the third mortality risk factor worldwide, with poor consumption of wholegrain cereals and high intake of salt being the most detrimental dietary risks. Curbing all these adverse trends is directly linked to our food choices. Under this context, the evaluation of the health and sustainability impacts of alternatives for e.g. animal-based food proteins are central aspects of this discussion.

There will not be a unique and fast-applicable solution for a general shift towards sustainable and healthy diets. A range of strategies and interventions will be necessary, making the transition through informed choices and efficiency-gains. For this process, integration of the social, economic, and environmental pillars of sustainability with the health risks and benefits associated with each food choice will be crucial, and a challenging and emerging area of research that is insufficiently addressed.

Risk-benefit assessment (RBA) of foods, a decision-support tool, estimates the overall human health impact following exposure (or lack of exposure) to a particular food or food component, integrating risks and benefits in comparable measures. It integrates knowledge on nutrition, toxicology, microbiology, chemistry and epidemiology for comprehensive health impact assessments. Methods that also consider the environmental impact of the food production systems are needed to perform a sustainable holistic assessment. Life cycle assessment (LCA) is the method typically recommended by international institutions to support policy making by quantitatively evaluating and assessing the effectiveness of sustainability interventions during the entire life cycle of a product. However, despite all the know-how generated in the last years, a framework that integrates simultaneously health risk-benefit and sustainability in a common framework is currently lacking.

“ALTERNATIVA | Alternative protein sources in the European diets – integrating health risk-benefit and sustainability” aimed to develop a holistic approach to assess the impact of alternative protein sources, integrating health and sustainability assessment (including health, environment, economy and societal aspects). Specifically, this project aimed to gather and combine expertise in risk-benefit assessment and sustainable impact assessment, to enhance the capacity of partners involved and to apply this to the evaluation of the impact of replacing red meat consumption with alternative protein sources. This project was conceived

considering three-pillars: i) capacity building, ii) tackle new global challenges, and iii) interaction with stakeholders and dissemination of outputs. The project was structured in five complementary work packages (WP).

Training activities, based on knowledge exchange, were organized to enhance the capacity of all partners involved in the two major aspects of the project i) risk-benefit assessment; and ii) sustainability assessment. The training activities were organized in two parts including the introduction to theoretical concepts and a practical session to explore typical cases. Brainstorming activities were also organized to harmonize languages among the different areas of expertise.

A protocol for integrating health (risk-benefit) assessment and environmental, economic, and social sustainability assessment was also developed. Transparent guidelines and objective criteria were considered to develop a harmonized approach considering multiple impacts and outputs from different disciplines and to communicate the results to the involved stakeholders. The protocol allows the identification of the case study and the most relevant risk-benefit and sustainability indicators. It shows the methodology options for data collection and assessment underlying scenarios analysis, as well as options to communicate results to facilitate the decision-making process for stakeholders. This protocol incorporates scientific and policy-relevant issues and is divided into three main parts: Part A – Initial Setting: Definition of the case study and indicators; Part B – Implementation: Methodology options for data collection and assessment; and Part C – Integration and Communication: Analysis and Communication.

Additionally, the overall impact of different possibilities of substitution of red meat with vegetable alternatives, pulses, in Portugal and Denmark was also investigated. The methodology described in the protocol was applied to this case study. The health and sustainability impact of different scenarios of substitution of red meat by pulses in the Portuguese and Danish populations was estimated. Multi-criteria Decision Analysis (MCDA) was developed by involving experts from the different domains who met in multiple meetings to develop the value tree and define criteria within each dimension. A positive health impact of all scenarios of the dietary substitution on the two populations was estimated. The impact of reducing the risk of colorectal cancer (CRC), diabetes, and ischaemic heart disease (IHD) because of reducing beef and increasing pulses consumption was higher for each increase in substitution. The estimated positive impact was larger for the Danish population than for the Portuguese. It varied between 5.9 DALYs (Disability-adjusted life years) (25% substitution) and 20 DALYs per 100,000 prevented (100% substitution) in Portugal. In Denmark, the maximum health gain would also be observed with 100% substitution of beef by pulses, with an estimated 39.7 DALYs per 100,000 prevented; 25% substitution would lead to 10.7 years of life saved per 100,000. The reduction of the risk of CRC was the largest contribution to the positive health effect in Denmark in all scenarios, whereas prevention of IHD had the largest impact in Portugal. The sustainability assessment showed positive economic impacts of the substitutions in the two countries. Beef production seems more resilient and stable than pulses in the two countries. The measured environmental impacts of the beef value chain were consistently higher in the two countries. Beef production was estimated to have a higher contribution to environmental degradation and resource use, with greenhouse gas (GHG)

emissions, eutrophic emissions, and land use showing the greatest differences between the two food groups.

In parallel and in order to assess competencies questionnaires to assess individual capacities within an Organization's staff and one to be filled at the level of the Organization were developed. Risk assessors, risk managers and communicators participated in two rounds of discussions which aimed to gain insight into relevant topics that could be included in the questionnaire (round 1) and discuss the content and structure of the draft version of the questionnaire (round 2), following suggestions collected in round 1. Assessors, managers and communicators from European national food safety authorities, the European Food Safety Authority (EFSA) scientific staff and experts, the EFSA focal points, the International RBA and Sustainability Networks, and representatives from FAO and WHO were invited to participate in the discussions. Personnel from national food safety authorities of countries participating in the ALTERNATIVA project together with EFSA staff and experts with possible knowledge and/or a special interest in performing RBA in the food sector finally accepted to participate in the discussions. A questionnaire designed to collect information on the current understanding and attitude towards a risk-benefit assessment on health and sustainability aspects was developed.

Overall, the ALTERNATIVA project has advanced the already established methodologies for assessing the health and sustainability impacts of food. In addition, the project provided evidence on the effects of important changes in the consumption patterns of certain foods that are traditionally associated with greater health and environmental effects. Overall, the results have the potential to open up new avenues of research that will benefit humans, animals and the environment.

## Table of contents

|                                                                                                                                       |    |
|---------------------------------------------------------------------------------------------------------------------------------------|----|
| Abstract .....                                                                                                                        | 1  |
| Summary .....                                                                                                                         | 4  |
| 1 Introduction .....                                                                                                                  | 9  |
| 1.1 Background and terms of reference as provided by the requestor .....                                                              | 11 |
| 2 Data and Methodologies.....                                                                                                         | 15 |
| 2.1 General approach to the project.....                                                                                              | 15 |
| 2.2 Training activities.....                                                                                                          | 16 |
| 2.3 Protocol of integration of risk-benefit assessment and sustainability .....                                                       | 16 |
| 2.4 Selection of food to be considered in the case study.....                                                                         | 17 |
| 2.5 Case study: replacing red meat consumption by pulses in two countries,<br>Portugal and Denmark .....                              | 18 |
| 2.5.1 Social impact assessment .....                                                                                                  | 18 |
| 2.5.2 Economic impact assessment .....                                                                                                | 19 |
| 2.5.3 Environmental impact assessment.....                                                                                            | 20 |
| 2.5.4 Health impact assessment.....                                                                                                   | 20 |
| 2.5.5 Multi-criteria decision analysis .....                                                                                          | 21 |
| 2.6 Stakeholders' current understanding and attitudes towards a risk-benefit<br>assessment on health and sustainability aspects ..... | 22 |
| 2.6.1 Assessing stakeholders' current understanding and attitudes towards a RBA on<br>health and sustainability aspects.....          | 22 |
| 3 Milestones achieved and results obtained .....                                                                                      | 25 |
| 3.1 Kick-off and final meeting .....                                                                                                  | 25 |
| 3.2 Action plan.....                                                                                                                  | 25 |
| 3.3 Training activities.....                                                                                                          | 26 |
| 3.4 Protocol of integration of risk-benefit assessment and sustainability .....                                                       | 26 |
| 3.4.1 Part A – Initial Setting: Definition of the case study and indicators.....                                                      | 26 |
| 3.4.2 Part B – Implementation: Methodology options for data collection and<br>assessment.....                                         | 27 |
| 3.4.3 Part C – Integration: Analysis and Communication.....                                                                           | 28 |
| 3.5 Case study: replacing red meat consumption by pulses in two countries,<br>Portugal and Denmark .....                              | 31 |
| 3.5.1 Social impact assessment .....                                                                                                  | 31 |
| 3.5.2 Economic impact assessment .....                                                                                                | 34 |
| 3.5.3 Environmental impact assessment.....                                                                                            | 35 |
| 3.5.4 Health impact assessment.....                                                                                                   | 37 |

|       |                                                                                                                                |    |
|-------|--------------------------------------------------------------------------------------------------------------------------------|----|
| 3.5.5 | Multi-criteria decision analysis .....                                                                                         | 40 |
| 3.6   | Stakeholders' current understanding and attitudes towards a risk-benefit assessment on health and sustainability aspects ..... | 44 |
| 3.6.1 | Main inputs of the first round of stakeholder consultations .....                                                              | 44 |
| 3.6.2 | Main inputs of the second round of stakeholder consultations .....                                                             | 46 |
| 3.7   | Communication: main outputs, policy brief and web-assisted international meeting .....                                         | 46 |
| 4     | Conclusion .....                                                                                                               | 48 |
| 4.1   | Impact assessment .....                                                                                                        | 49 |
| 5     | Recommendations .....                                                                                                          | 52 |
|       | References .....                                                                                                               | 53 |
|       | Abbreviations.....                                                                                                             | 58 |
|       | ANNEXES (6) .....                                                                                                              | 59 |

# 1 Introduction

The pressure global food systems exert to the environment has amplified over time, following growth of the human population, changes in the food consumption patterns and the intensification of the production systems (Fasolin et al., 2019). By 2050, a growth of more than 30% of the current global population is expected (Fasolin et al., 2019; Tian et al., 2016). This increase will be followed by an increase in demand for food (Aiking & de Boer, 2018). Although technological innovation contributed to increase food availability, security and safety with a consequential reduction in hunger, poverty and mortality worldwide, the agri-food system pressure on the environment has amplified over time. Therefore, without virtuous transformations, an overpressure on agriculture and food production will be unavoidable, with detrimental consequences for the environment and human health. Food production, primarily agriculture, causes negative externalities, such as the contribution to climate change, exploitation and pollution of land and water bodies, which are among the largest drivers of biodiversity loss (Fanzo & Davis, 2019; Joyce, Dixon, Comfort, & Hallett, 2012). In particular, animal-derived protein production like ruminant meat and dairy contribute at most to the environmental impact (Fasolin et al., 2019; Nadathur et al., 2017; Spector & Gardner, 2020; van der Spiegel et al., 2013). From the consumption side, unhealthy diets are the third mortality risk factor worldwide, with poor consumption of wholegrain cereals and high intake of salt being the most detrimental dietary risks. Curbing all these adverse trends is directly linked to our food choices (Willett et al., 2019).

Even though a growing number of international organizations and governments have recognized that food policies should aim to integrate the dual objective of promoting human health and protect the environment, integrated strategies are still largely lacking (Gonzalez Fischer & Garnett, 2016; Smetana et al., 2019).

Committed to build a better future for humankind and recognizing the immense challenges to sustainable development, the United Nations (UN) adopted the 17 Sustainable Development Goals (SDGs) of the 2030 Agenda for Sustainable Development (United Nations, 2015), offering a vision for food and agriculture as a key to sustainable development. In the European Union (EU), food production still results in air, water and soil pollution, impacts climate change, and consumes excessive amounts of natural resources, with an important part of food waste (European Commission, 2019). As a growth strategy, the European Commission established a set of policy initiatives with the overarching aim of making Europe climate neutral in 2050, named *The European Green Deal*. One of the policy actions is "From Farm to Fork" and considers the development of more sustainable food systems as a priority. It recognizes that food production needs to evolve to sustainable exploitation of natural resources and simultaneously meet the growing demand for a balanced diet focused on healthier products (Smetana et al., 2019; van der Goot et al., 2016). As such, the future needs sustainable food systems that deliver foods for all in a way that optimal economic, social, health and environmental outcomes are ensured and sustained (Fasolin et al., 2019; Gonzalez Fischer & Garnett, 2016; Smetana et al., 2019).

Shifts towards sustainable and healthier diets are an urgent and effective way for driving the changes that are expected and urgently needed. To promote such changes and support transformations of current food systems and food consumption choices, policymakers need evidence on the integrated impacts of different options of dietary shifts. These assessments can be made through the creation of sustainable and healthy roadmaps, and by applying multi and interdisciplinary approaches and engaging a broad range of stakeholders with different but complementary views and values (Ambikapathi et al., 2022).

Under this context, the evaluation of health and sustainability impacts of alternatives for e.g. animal-based food proteins are central aspects of this discussion (Aiking & de Boer, 2018). Consumers appear to be increasingly interested in sustainable protein sources that can meet their nutritional and sensory needs (Henchion et al., 2017). Vegetable protein sources seem to be the natural alternative to animal-based proteins, since some of them are already present in the current human diet (Fasolin et al., 2019; Henchion et al., 2017). These include grains (e.g., wheat and maize), seeds (e.g., chia), leaves (e.g., moringa) and pulses (e.g., beans, lentils). In addition to lower GHG emissions, less resource-intensive and less environmentally destructive production than animal husbandry, and lower production associated costs, plant-based diets are usually linked to lower risk for developing chronic diseases (e.g. ischemic heart disease, cancer, Type 2 Diabetes or metabolic syndrome). These health benefits may be associated with their content of numerous health promoting nutrients (e.g. vitamins, minerals, fibers), antioxidants and anti-inflammatory agents that are usually present in these food products (Lynch et al., 2018). However, there are also concerns related to exclusively or almost exclusively vegetarian diets such as vegan or diets containing plant proteins as the major source of protein: they may lead to nutrient deficiency due to unbalanced protein contents (Henchion et al., 2017). Plant proteins may not have all the essential amino acids in the required proportion by humans, when considered as single foods. They may also contain anti-nutritional compounds that could inhibit some biological functions (Henchion et al., 2017; Lonnie et al., 2018; Lynch et al., 2018). Lastly, vegetarian diets and diets high in plant-based material may lead to higher exposures to pesticides, environmental contaminants and natural toxins frequently present in these foods. Other examples of alternatives to animal-derived proteins are edible insects, mycoproteins, proteins from microalgae.

There will not be a unique and fast-applicable solution for a general shift towards sustainable and healthy diets. A range of strategies and interventions will be necessary, making the transition through informed choices and efficiency-gains. For this process, integration of the social, economic, and environmental pillars of sustainability with the health risks and benefits associated with each food choice will be crucial, and a challenging and emerging area of research that is insufficiently addressed (Clark et al., 2019; Hollander et al., 2019; Springmann et al., 2018).

Risk-benefit assessment (RBA) of foods, a decision-support tool, estimates the overall human health impact following exposure (or lack of exposure) to a particular food or food component, integrating risks and benefits in comparable measures. It integrates knowledge on nutrition, toxicology, microbiology, chemistry and epidemiology for comprehensive health impact assessments (Assunção et al., 2019; Boué et al., 2015; Naska et al., 2022; Pires et al., 2019). RBA is of interest of authorities from food-related areas to develop food policy and consumer

advice, to businesses developing new food products, and to consumers considering dietary changes (Hoekstra et al., 2013).

Methods that also take into account the environmental impact of the food production systems are needed to perform a sustainable holistic assessment. Life cycle assessment (LCA) is the method typically recommended by international institutions, such as the European Commission and the United Nations Environment Programme, to support policy making by quantitatively evaluating and assessing the effectiveness of sustainability interventions during the entire life cycle of a product (Cucurachi et al., 2019; Sala et al., 2017; Takacs & Borrión, 2020). In parallel, the outputs mainly produced by European Projects (e.g., GLAMUR<sup>1</sup>; and Strength2Food<sup>2</sup>) have also contributed significantly to advance the knowledge and the tools to assess the sustainability impact of food choices.

However, despite all the know-how generated in the last years, a framework that integrates simultaneously health risk-benefit and sustainability in a common framework is currently lacking.

## 1.1 Background and terms of reference as provided by the requestor

The grant *GP/EFSA/ENCO/2020/03 | Partnering grants – GA 2* corresponding to the project *ALTERNATIVA | Alternative protein sources in the European diets – integrating health risk-benefit and sustainability* was awarded by EFSA to a consortium composed by six European institutions, including:

Instituto Nacional de Saúde Doutor Ricardo Jorge, I.P (INSA), the coordinator of the Consortium, established in Portugal;

French National Institute for Agriculture, Food and Environment (INRAE), established in France;

Danmarks Tekniske Universitet (DTU), established in Denmark;

Εθνικό και Καποδιστριακό Πανεπιστήμιο Αθηνών (NKUA), established in Greece;

University of Parma (UNIPR), established in Italy;

Autoritatea Națională Sanitară Veterinară și pentru Siguranța Alimentelor (ANSVSA), established in Romania.

Note: The Romanian partner was able to contribute and collaborate only in the project initiation phase that required later adjustments during the project implementation to achieve the set project objectives.

ALTERNATIVA aimed to develop a holistic approach to assess the impact of alternative protein sources, integrating health and sustainability assessment (including health, environment,

<sup>1</sup> <https://cordis.europa.eu/project/id/311778/it>

<sup>2</sup> <https://cordis.europa.eu/project/id/678024>  
[www.efsa.europa.eu/publications](http://www.efsa.europa.eu/publications)

economy and societal aspects). Specifically, this project aimed to gather and combine expertise in risk-benefit assessment and sustainable impact assessment, to enhance the capacity of partners involved and to apply this to the evaluation of the impact of replacing red meat consumption by alternative protein sources.

To achieve this aim, ALTERNATIVA had the following specific objectives:

- To enhance capacity-building among members of the project team in applying methodologies related to the health RBA and sustainability impact assessment;
- To develop, in collaboration with different stakeholders (including EFSA), a protocol to integrate the assessment of the impact of foods/diets on health and the impact on sustainability;
- To apply the derived framework to a case study, to validate it with a real case scenario, estimating the health and sustainability impacts of replacing red meat consumption by alternative protein sources; and
- To disseminate and promote the established method of integrated health and sustainability assessment to other potential EU users.

ALTERNATIVA was developed by a multidisciplinary team of partners from national health and food institutes, national food authorities, R&D institutions and academia. Six institutions from six different countries (Portugal, France, Denmark, Greece, Italy and Romania) integrated the project's consortium. The partners were geographically spread across the EU, provided expertise on the different disciplines of the project (risk-benefit assessment and sustainability impact assessment), and had different roles as food safety stakeholders. The complementary scientific expertise needed to address the aims of ALTERNATIVA, including nutrition, microbiology, toxicology, epidemiology, economics, environmental sciences, were covered by the multi-disciplinary consortium.

The National Institute of Health Dr. Ricardo Jorge (INSA) from Portugal was the coordinator of the project. INSA develops its activity as a State Laboratory of the health sector, National Reference Laboratory and National Health Observatory. The Food and Nutrition Department (DAN) of INSA develops activities in the areas of food safety and nutrition aiming to contribute to public health gains through the study of the situation of the country in the scope of food safety and nutrition. The department works in the promotion of health, prevention of food-borne diseases and improvement of the nutritional status of the Portuguese population, developing activities of research, surveillance, reference, providing a diversity of services, training, information, and consultation. Additionally, regarding risk-benefit assessment, INSA has been developing several activities, including the coordination of a recent EFSA's Partnering Grant, as the RiskBenefit4EU project (AGREEMENT NUMBER – GA/EFSA/AFSCO/2017/01 – GA02).

INRAE is the French National Institute for Agriculture, Food and Environment under the dual aegis of the Ministry of Research and the Ministry of Agriculture. Its scientific community of 12 000 people located in 250 different units are dedicated to research, innovation, and support for public policies to guide the emergence of sustainable agricultural and food production systems. The INRAE-ONIRIS UMR1014 SECALIM laboratory is a joined research

unit depending also on the Nantes-Atlantic National College of Veterinary Medicine, Food Science and Engineering (ONIRIS). Composed of 30 people, INRAE-SECALIM develops research in food safety and quality to progress on the study of quantitative risk assessment to control pathogenic and spoilage microflora of food and the microbial ecology of food products using molecular tools. Since 2014, they developed new activities in the emerging area of RBA to develop quantitative models assessing the overall impact of food on human health considering microbiological, chemical, and nutritional dimensions. They have so far accomplished two PhD in this area and contributed to two EFSA Partnering Grants, including RiskBenefit4EU (GA/EFSA/AFSCO/2017/01 – GA02) and NovRBA (GP/EFSA/ENCO/2018/03 – GA01).

The National Food Institute, Technical University of Denmark (DTU Food) researches and communicates sustainable and value-adding solutions in the areas of food and health for the benefit of society. DTU Food has a long tradition of advice-giving to national and international food and public health authorities and of research in nutrition, epidemiology, toxicology and microbiology. It is now a leading institution in the development and application of risk-benefit assessment methods. It leads the International Risk-Benefit Assessment Network and coordinates and participates in several risk-benefit assessment projects.

The National and Kapodistrian University of Athens (NKUA) is among the largest universities in Europe with 33 Departments about 62.700 students (undergraduate, postgraduate and doctoral) and 5.448 foreign students. The Department of Hygiene, Epidemiology and Medical Statistics (DHMS) which was involved in this research proposal comprises a faculty of 13, supporting faculty of three and a staff of eight administrators, IT specialists and food technologists. Several research associates and graduate students also work in the Department. The DHMS is responsible for the teaching of Epidemiology-Research Methodology, Preventive Medicine-Public Health and Biostatistics to undergraduate students at the School of Medicine and the Departments of Pharmacy, Dentistry and Biology of NKUA. It also leads or participates in the series of postgraduate courses organized by NKUA, some of them in collaboration with other academic institutions in Greece. The Department has a strong presence in research with participation in international projects and several of its faculty have strong publication and citation records. Research interests cover the fields of nutritional epidemiology, chronic and infectious disease epidemiology and biostatistics. The DHMS was coordinating the NovRBA project (GP/EFSA/ENCO/2018/03 – GA01). Furthermore, faculty members have long served as Coordinators/Members of national Committees responsible for policy formulation and strategy development of the Ministries of Health, Agriculture, Research and Development.

The University of Parma (UNIPR) brought together a strong interdisciplinary team comprising agricultural economists based at the Department of Economics; Food and Drug; and Biosciences. This team has substantial research experience in the assessment and evaluation of the economic, social and environmental sustainability of agribusiness systems (e.g., quality food products), supply chain management systems (e.g., local vs global food chains), agricultural policies, as well as consumer choice and dietary behaviour. The social, environmental and economic sustainability of food products, food chains and dietary behaviours have been assessed in other EU projects, e.g., STRENGTH2FOOD (Strengthening European Food Chain Sustainability by Quality and Procurement Policy, Horizon 2020,

<https://www.strength2food.eu>, 2016-ongoing), GLAMUR (Global and Local food chain assessment: a multidimensional performance-based approach, 7th F.P., <http://glamur.eu>, 2014-2016), and PRIMEFISH (Developing Innovative Market Orientated Prediction Toolbox to Strengthen the Economic Sustainability and Competitiveness of European Seafood on Local and Global markets, Horizon 2020, <http://www.primefish.eu>, 2015-2019). The consumers acceptance of novel foods containing whole or processed insects has been also widely investigated by the UNIPR team (Mancini et al., 2019; Davide Menozzi et al., 2017; Sogari et al., 2019).

The National Sanitary Veterinary and Food Safety Authority (ANSVSA) in Romania is the central competent sanitary veterinary and food safety authority that coordinates, technically and administratively, the national veterinary and food safety services, and also control the public veterinary and food safety activities. The ANSVSA exerts the following functions: i) Regulation, by which it ensures that the legal and specific sanitary veterinary and food safety framework is compliant with the national and international provisions; ii) Control, by which it enforces the surveillance and control of the enforcement of the specific legal provisions; iii) Strategy, by which it elaborates, according to the national and international policies, the sanitary veterinary and food safety strategy, and also the risk evaluation, analysis, management and communication and the research activities in its area of expertise; iv) Administration, by which it coordinates the subordinated structures and ensures the management of the national veterinary and food safety services; v) Representation, by which it ensures the internal and external representation of the Romanian veterinary services.

ALTERNATIVA planned to expand and progress on dimensions usually addressed separately, i.e. health risk-benefit assessment and sustainability impact assessment, applying a harmonized approach that facilitates the integration of outputs and the communication of results to the involved stakeholders.

## 2 Data and Methodologies

### 2.1 General approach to the project

Under the ALTERNATIVA project, as detailed before, a multidisciplinary team of partners from national health and food institutes, national food authorities, R&D institutions and academia was gathered.

This project was conceived considering three-pillars: i) capacity building, ii) tackle new global challenges, and iii) interaction with stakeholders and dissemination of outputs, and was structured in five work packages (WP) (Figure 1).

WP 1 aimed to i) coordinate all planned activities under ALTERNATIVA, including project management timely progress; ii) manage financial issues related with the project; iii) disseminate overall results; and iv) perform the global evaluation and final report of the project. Specific tasks included: i) organization of internal and external (with EFSA) meetings; ii) monitoring and evaluation of work progress and project achievements; iii) interpretation and integration of the obtained results; and iv) planning subsequent activities and publications.

WP 2 aimed to harmonize the knowledge and enhance the capacity of all partners involved on the methodologies of RBA and sustainability impact assessment, through training activities. The contents of these training sessions comprised two major aspects: i) risk-benefit assessment; and ii) sustainability assessment.

WP 3 aimed to develop a coherent framework expanding the current RBA approaches with measures of the sustainability of food products (i.e., environmental, economic and social impacts). This framework was developed to support the case study (WP 4), concerning the impact of substituting red meat consumption by pulses.

WP4 aimed to apply the framework developed in WP 3 and implement it in a real-life case study. The case study tested the developed innovative methodology to assess the health and sustainability effects of red meat substitution by pulses.



Figure 1. Schematic organization of the ALTERNATIVA work packages.

WP 5 aimed to gain insight on the current understanding, interests, needs and questions of relevant stakeholders regarding the integration of RBA and sustainability impacts' assessment in their practice - foreseeing thus the global challenges generated by the current shift to sustainable food systems and diets. This WP had a transversal dimension, contributing to bridge the technical/scientific aspects of the project with the real demand of the policymakers, accumulating the expertise generated through WP2 (training and capacity building) and WP4 (case-study) into a single portfolio.

## 2.2 Training activities

Training activities, based on knowledge exchange, were organized to enhance the capacity of all partners involved on the two major aspects of the project i) risk-benefit assessment; and ii) sustainability assessment. It was organized in two parts including the introduction to theoretical concepts and a practical session to explore typical cases. Brainstorming activities were also organized to harmonize languages among the different area of expertise. A total of four virtual seminars of 2h00 to 3h00 were organized and recorded, involving project partners and invited experts.

The RBA-focused seminar covered: the stepwise approach for RBA, the definition of a RBA case study, the identification and selection of health effects (following the strategy developed in the 64/134 17 EFSA-supported NovRBA project), dose response in nutrition, toxicology and microbiology, quantification of health impacts, and managing deterministic and stochastic data. The second seminar on sustainability assessment focused on: definition of key terms, scoping the assessment, general framework to assess environmental, social and economic impacts, and introduction to the available tools. Both seminars used practical examples to illustrate different concepts and methods. After completion the two theoretical seminars, plenary brainstorming sessions were organized to generate ideas of strategies to integrate them into a common framework.

## 2.3 Protocol of integration of risk-benefit assessment and sustainability

The partner consortium has developed a protocol integrating health (risk-benefit) assessment and environmental, economic, and social sustainability assessment. Transparent guidelines and objective criteria are applied to develop a harmonized approach considering multiple impacts and outputs from different disciplines and to communicate the results to the involved stakeholders. The protocol allows the identification of the case study and the most relevant risk-benefit and sustainability indicators. It shows the methodology options for data collection and assessment underlying scenarios analysis, as well as options to communicate results to facilitate the decision-making process for stakeholders.

The protocol has been developed based on consultations between researchers, scientific experts, and other relevant stakeholders (e.g., policy makers). An expert group of 27

scientists from different Universities, research institutes and national authorities was established to cover expertise in the different dimensions of sustainability (social, environmental, and economic) and risk-benefit assessment (RBA). The expert group was involved in all the steps during the establishment of the protocol.

The protocol has been designed in such a way it has a wide application; it should be applied collaboratively, to ensure the best availability of information and points of view. Policy and other decision-makers were involved in providing feedback and comments on the protocol<sup>3</sup>, as well as in participating directly in the case study application in the considered different phases. They were involved in discussing the definition of the case study, evaluating the set of indicators, and discussing the outcome of the study. The three main dimensions of sustainability are considered (i.e., the economic, environmental, and social dimensions), and the fourth health dimension has been added to integrate the RBA in the integrated assessment.

The sustainability assessment of agri-food products performance is mainly based on the SAFA-Sustainability-Guidelines (Sustainability Assessment of Food and Agriculture systems) from the Food and Agriculture Organisation of the United Nations (FAO) (FAO, 2013a). The SAFA guidelines have been produced as a code of practice, or guidelines, to assist in achieving sustainable and fair practices in food and agriculture production and trade.

The health impact assessment is based on risk-benefit assessment (RBA) of foods methodology, a well-established decision-support tool that integrates the assessment of beneficial and adverse health effects of foods, food components and diets (Assunção et al., 2019; Pires et al., 2019). While there are different approaches for RBA, we have selected the one that estimates health effects using the population health metric “disability-adjusted life year” (DALY). DALY, a composite health metric that considers both morbidity and mortality, is the most commonly used health indicator for quantitative RBA. In this context, one DALY represents one year of healthy life lost.

## 2.4 Selection of food to be considered in the case study

As part of this project, we planned to apply the developed protocol of integration to a specific case study. As planned in the proposal, this case study must consider the substitution of red meat consumption with an alternative protein source. Therefore, a reflection process was implemented between the experts in the team, focusing on the pros and cons of the different alternatives, i.e. insects and pulses. Through a guided brainstorming process, each team member was able to identify their arguments and present them to the whole team. A voting process was then considered, and a decision was made. Pulses were chosen because: the food category is well established in European diets; the research question is pertinent to

<sup>3</sup> In particular, the project’s Consortium decided to rely on stakeholder consultations for the development of a questionnaire to assess how prepared stakeholders are for the application of a food-related risk-benefit assessment framework addressing sustainability aspects with a special focus on health. Two rounds of focus group discussions have been conducted involving risk managers and communicators, and risk assessors to gain in-depth insight around relevant topics and discuss the content and structure of the draft version of the questionnaire.

[www.efsa.europa.eu/publications](http://www.efsa.europa.eu/publications)

provide evidence for policy makers that could be used within a short time; it could be translated into food-based dietary guidelines; data for some variables is available for the two countries; and the intake of pulses has increased in recent years.

## 2.5 Case study: replacing red meat consumption by pulses in two countries, Portugal and Denmark

We investigated the overall impact of different possibilities of substitution of red meat with vegetable alternatives (pulses) in Denmark and Portugal. The case study was selected due to its relevance for policy strategies in the framework of achieving sustainable development goals (SDGs) specifically related to carbon footprint reduction (SDG 12 – Responsible consumption and production; SDG13 – Climate action), and public health objectives (SDG3 – Good health and well-being; SDG4 – Quality education; SDG8 – Decent work and economic growth), which are directly linked to food security and diet quality improvement (SDG2 – Zero hunger).

The methodology applied was developed to be applicable to this and other research questions and has been described in the protocol (see section 2.3).

We estimated the health impact of different scenarios of substitution of red meat by pulses in the Portuguese and Danish population in the current time, i.e., using the most recent available food consumption, disease incidence and demographics data. We defined “red meat” as unprocessed beef meat and excluded other meat types. We defined pulses as a mix of white beans, brown beans, lentils, chickpeas and peas. The contribution of the different types of pulses to consumption in all scenarios varied between the two countries, following the preferences of current consumption in the Portuguese and Danish population.

The scenarios compared were:

Baseline: current consumption of red meat and pulses

Alternative 1: Replacing of 25% of beef by proportional amounts (gram to gram) of pulses.

Alternative 2: Replacing of 50% of beef by proportional amounts of pulses.

Alternative 3: Replacing of 75% of beef by proportional amounts of pulses.

Alternative 4: Replacing of 100 % of beef by proportional amounts of pulses.

### 2.5.1 Social impact assessment

In the case of social impact assessment, a severe limitation relates to data availability, since the aspects worth to be investigated are rarely traced in national and international database and need to be collected directly. For the social pillar in the sustainability assessment, we rely on secondary data gathered from Eurostat and FAOStat database. The dimensions investigated are female labour force (general workforce and manager position) and

affordability. Qualitative aspects related to tradition and acceptance were included in the Multi-criteria Decision Analysis (MCDA).

From the Eurostat database we selected the following farm categories to include in the case study: the specialist cattle-rearing and fattening (code: FT46\_SO) and the specialist cereals, oilseed and protein crops (code: FT15\_SO). From the FAOStat database we gathered the cost of legumes, nuts, and seeds. The cost of red meat was obtained from the website [globalproductprices.com](http://globalproductprices.com), since the basket of animal sources considered in the FAOStat underestimates the sole cost of meat.

We considered the women directly employed by the farm on a regular basis for assessing the gender equality in labour market (from the EF\_LF\_SIZE dataset). Regarding the farm managers, both the women directors of group holding (dataset EF\_M\_FARMANG) and the managers of no-group holding (dataset EF\_LF\_SIZE) have been considered.

The substitution scenarios for the social indicators associated to female employment were assessed by assuming an increase in the production of pulses with the consequent complementary decrease of meat production, which, in turn, were related with the necessary annual working unit (AWU) in the two sectors. In other words, no conversion index to pair energy balance or any other intake balance was adopted. The hypothesis is that a reduced consumption will lead to a fall in the production. We estimated results by computing on the total yearly production, gathered by the Eurostat database coherently with the same category selected for the labour market. Pulses data are with the crop production category (dataset APRO\_CPSH1), and for meat in the slaughtering (dataset APRO\_MT\_PANN).

The affordability index represented the cost of purchasing the least expensive locally-available food item to meet daily intake levels recommended in food-based dietary guidelines and is expressed in international dollars using purchasing power parity (PPP). The affordability index was scaled accordingly with the consumed quantities for the meat; for pulses it was estimated considering the composition of the Healthy Diet Basket proposed by the FAO.

## 2.5.2 Economic impact assessment

For assessing the economic impact of substituting, we focused on secondary data related to the financial performance of the farms, with information on prices, gross product, value-added, net margin. We considered the impact in terms of employment and local development too. Secondary data were collected from official data sources (e.g. Eurostat) at the national level when possible, from international scientific literature and relevant projects. If not, different sources were used from other countries, or from wider geographical contexts.

Income-based indicators (i.e. gross operating margin, GOM, and net result) were calculated as % of turnover for each target food (pulses) and its standard reference (beef). The second set of economic indicators is related to the resilience and stability of the market. First, we used a first set of indicators referred to the international trade: exported volumes and values, and the ratio of the products exported (in volume) to the total production, provide some information on international market dynamism (Arfini & Bellassen, 2019). For assessing the

stability of the market, we estimated the price volatility with the coefficient of variation (i.e. the ratio of mean and standard deviation of the prices).

Finally, we assessed the contribution to local sustainable economic development, indicating the capacity of the production system to contribute to local employment and to economic spill-over on the territory. First, we used the regional workforce in Denmark and in Portugal. Then, the economic spill-over indicator was applied to measure the return on the territory, i.e. the share of expenditure retained in the local economy. We applied a specific measure (i.e. the Local Multiplier – LM3 indicator) derived in aggregate terms at European level by Donati et al. (Donati et al., 2021), in general for vegetable and animal products.

### 2.5.3 Environmental impact assessment

To estimate the environmental impacts of alternative food consumption scenarios, multiple environmental impact indicators have been selected, including greenhouse gas (GHG) emissions (kg CO<sub>2</sub> eq), eutrophying emissions (g PO<sub>4</sub><sup>3-</sup>), water use (L), arable land use (m<sup>2</sup>\*year), pasture land use (m<sup>2</sup>\*year), total land use (by summing arable and pasture land use), and biodiversity impact (species destined to extinction per year).

To collect consistent data among the different indicators, the WWF dataset has been consulted as the main reference for quantitative data searching. More specifically, the environmental impacts considered are the “average” impacts referring to the current consumption, which are distinguished from the “marginal” impacts referring to the additional unit of production.

For all the indicators mentioned, by selecting Denmark and Portugal as countries of interest, the averages of the records referred to both “beans & pulses” and “peas” and to “bovine meat (beef herd)” and “bovine meat (dairy herd)” listed in the “Data – Product Level” sheet of the dataset have been calculated for the two selected “pulses” and “beef” categories, respectively. As the dataset uses 1 kg or 1 L of food products as functional unit, the data have been subsequently corrected to obtain the impacts referred to the serving size (i.e., 100 g).

As health-related environmental indicator, photochemical smog has been considered. In this case, the data have been retrieved from a LCA food database specifically referred to food products produced and consumed in Denmark (<http://www.lcafood.dk/>). Due to data availability issues, the ALTERNATIVA team decided to apply for Portugal the same emission factors as used for Denmark.

### 2.5.4 Health impact assessment

The assessment of the health impact of the different scenarios was based on risk-benefit assessment (RBA) methodology, a well-established decision-support tool that integrates the assessment of beneficial and adverse health effects of foods, food components and diets (Assunção et al., 2019; Pires et al., 2019). While there are different approaches for RBA, we selected the one that estimates health effects using the population health metric “disability-

adjusted life year” (DALY), the metric most often used as health indicator for quantitative RBA. The approach applied, including the identification of health outcomes, risk and benefit characterization, and DALY calculation, were based on (Fabricius et al., 2021). The health outcomes included in the model are presented in Table 1.

Table 1: Health outcomes included in the health impact assessment of different scenarios of substitution of red meat by pulses.

| Food            | Component   | Health outcome                           | Reference                           |
|-----------------|-------------|------------------------------------------|-------------------------------------|
| Beef            | Whole food  | Increased risk of colorectal cancer      | (Norat et al., 2017)                |
| Beef            | Whole food  | Increased risk of diabetes mellitus      | (Pan et al., 2011)                  |
| Pulses          | Whole food  | Decreased risk of ischemic heart disease | (Afshin et al., 2014)               |
| Beef            | Vitamin B12 | Decreased risk of megaloblastic anaemia  | (Nordic Council of Ministers, 2014) |
| Beef and pulses | Iron        | Decreased risk of microcytic anaemia     | (Nordic Council of Ministers, 2014) |

We used data from the National Food, Nutrition and Physical Activity Survey of the Portuguese Population (IAN-AF), 2015–2016 (Lopes et al., 2018) and the Danish National Survey of Diet and Physical Activity (DANSDA), 2011–2013 (Pedersen et al., 2015) to model the substitution, food composition data from national databases and dose response data collected from the literature. We estimated incidence, mortality and DALYs for each health outcome in the different scenarios using probabilistic modelling.

### 2.5.5 Multi-criteria decision analysis

To estimate the integrated impacts of each scenario, we applied a Multi-Criteria Decision Analysis (MCDA) approach, which is an extension of decision theory that covers (decision) problems with increased complexity that is generally due to multiple, sometimes conflicting, objectives that are valued differently by different stakeholders. The core idea is to appraise options (or alternatives) on individual objectives (criteria) and then combine this partial appraisal into one overall appraisal. Before the modelling process began, steps were taken to identify experts that could act as representatives for each of the dimensions, with the objective being to make sure that every dimension is fairly represented in the overall discussion. Additionally, there was an effort to have at least two experts for each dimension. Because the project’s consortium included individuals with expertise in most dimensions, we opted by selecting experts within the group. We then approached experts within the fields for which participants were lacking, primarily focusing on specialists from the institutions involved in the consortium.

The first step in developing the model for the case-study was to identify the relevant criteria with which to evaluate the alternative scenarios. Each evaluation criterion is an attempt at representing a key concern for the stakeholders. To identify the criteria, a brainstorming session was held with each expert t. Then, there was an open discussion session, in which

concerns were grouped and adjusted until the remaining criteria were considered to meet the necessary conditions: be intelligible; be consensual; be isolable; be operational. After all rounds of selection of criteria, experts developed a “final value tree” (Figure 11) describing all criteria under each dimension (i.e., economic, social, health and environment) and assigned values and weights to each criteria based on the results of sections 2.4.1 to 2.4.5..

## 2.6 Stakeholders’ current understanding and attitudes towards a risk-benefit assessment on health and sustainability aspects

WP5 aimed to provide insight into the operationalization of the project’s scientific and technical aspects. It was designed to serve a dual role: to exploit the outcomes of WP2 (training and capacity building); to liaise with the development of the protocol (WP3) and with the case-study (WP4) to enhance their applicability and contribute to bridging technical and scientific features with the demand of managers and policymakers. To accomplish these aims, WP5 has been organized into two sub-tasks: (a) assessing stakeholders’ understanding and attitudes towards an RBA on health and sustainability aspects and (b) describe in a policy brief (Annex 3) how RBA methods can be put into practice, incorporating health and sustainability in an EU MS.

### 2.6.1 Assessing stakeholders’ current understanding and attitudes towards a RBA on health and sustainability aspects

The training seminars of WP2 enriched and aligned the Consortium’s knowledge and understanding of parameters that an RBA addressing sustainability multi-dimensionally should include. On the grounds of this common understanding, the literature was reviewed to identify consultations, surveys or questionnaires applied to assess stakeholders’ perceptions on food safety and risk assessment. Although some (limited) tools to map the situation were retrieved from the literature (Aldaya et al., 2021; Hollander et al., 2019; Malapit et al., 2017; Membré et al., 2021; van Kleef et al., 2006), a comprehensive questionnaire addressing all aspects was missing. Relevant literature proposed by project participants covering nutrition, microbiology, toxicology and sustainability assessment was further screened to identify aspects that should be addressed through the newly developed questionnaires.

The Consortium additionally contacted members of the International RBA Network and experts in the sustainability sector. In particular, invitation letters were disseminated to experts in the field in order to identify current knowledge on: (a) data sources (b) commonly used metrics for quantification of health effects, (c) concepts (e.g. epidemiology) and terminology (e.g. profiling of a food item) relevant for RBA, (d) opinions regarding the usefulness and applicability of RBA to support policy and (e) challenges encountered in RBA (development, application and interpretation of the results, including associated uncertainties).

To develop a questionnaire from the ground up, project participants decided to rely on crowd sourcing, undertaking stakeholder consultations through semi-structured interviews. Risk assessors, risk managers and communicators participated in two rounds of discussions which aimed to gain insight into relevant topics that could be included in the questionnaire (round 1) and discuss the content and structure of the draft version of the questionnaire (round 2), following suggestions collected in round 1. Assessors, managers and communicators from European national food safety authorities, the European Food Safety Authority (EFSA) scientific staff and experts, the EFSA focal points, the International RBA and Sustainability Networks, representatives from FAO and WHO were invited to participate in the discussions. Personnel from national food safety authorities of countries participating in the ALTERNATIVA project together with EFSA staff and experts with possible knowledge and/or a special interest in performing RBA in the food sector finally accepted to participate in the discussions.

The consultations were held virtually, after ensuring participants' consent and agreement, and followed a focus-group approach for the exchange of views and proposals through two rounds which shaped the final deliverables. Prior to the consultations, a short inventory regarding affiliations, field of expertise and familiarity with RBA methods was disseminated. The discussions were prepared and coordinated by a social psychologist with expertise in cross-cultural psychology and knowledge of methods around qualitative research. The methodology applied included:

(a) definition of the study sample. The list of potential focus group participants was selected through a sampling procedure to ensure appropriate coverage of the field and representativeness of different expertise.

(b) several preparatory meetings between the NKUA team and members of the ALTERNATIVA Consortium to formulate the script of discussions to be held in round 1 and discuss the procedure, particularly in relation to prioritizing key questions. The script was developing in parallel with the protocol for an integrated Risk-Benefit and Sustainability Assessment of foods. Hence, excerpts of the protocol were integrated in the script and focus group participants became acquainted with the methodological steps outlined in the protocol. Comments collected through the first round of discussion were also communicated to the University of Parma (leading the development of deliverable D3) and were considered in the preparation of the protocol assuring the necessary inter-relations between the project's WPs.

To guide discussions during round 1, the script included three sections with questions addressing the knowledge, perceptions and challenges foreseen in the application of RBA addressing health and sustainability aspects. In particular, Section A included questions on stakeholders' knowledge and prior experience in RBA methods integrating health and sustainability in food related matters. Section B aimed to assess stakeholders' expectations in relation to an RBA scoping questionnaire. Focus group participants were asked to describe fundamental aspects of health RBA and sustainability impact assessment that should be well established and clarified for the initiation and implementation of an integrated RBA framework. Questions about their perspective of the most relevant context for the application of an RBA scoping questionnaire (at individual or organizational level) and of the knowledge and expertise required were also considered during the discussion. This final section (Section C) aspired to investigate the challenges stakeholders foresee in the development and

application of such an integrated framework in relation to the availability of data and team competences (scientific background and expertise). Finally, experts were requested to discuss the associated uncertainties and their potential impact on results and conclusions based on their field of expertise (risk assessment, risk communication, risk management).

Based on knowledge gained through round 1, the project Consortium concluded on the necessity to develop two questionnaires: one to be filled in at the level of organization or any related Corporation and a second to assess individual capacities within an Organization's staff. The questionnaires aim to assess how prepared stakeholders are for the application of a food-related RBA framework addressing sustainability aspects with a special focus on health. They record and evaluate how risk assessors/managers and communicators understand RBA including sustainability and health dimensions; assess attitudes towards this approach; and, identify the sector's gap of knowledge so as to foresee barriers in their application at the individual and organizational level.

During the second round, participants reviewed preliminary versions of these questionnaires, assessed the clarity of questions, and discussed answers together with possible entries to be added. All comments were integrated into the final questionnaires.

## 3 Milestones achieved and results obtained

### 3.1 Kick-off and final meeting

As planned in the project proposal, kick-off and final meetings were organized to plan and discuss the activities to be performed under the ALTERNATIVA project, and the activities and main outputs obtained during the project, respectively.

The kick-off meeting occurred on 25<sup>th</sup> of June 2021, online, and considered the following topics for the agenda:

- 1) Welcome and opening
- 2) Tour de table
- 3) Adoption of the Agenda
- 4) Financial aspects and timelines on the administration of the project (presentation from EFSA side)
- 5) Scientific aspects (presentation from EFSA side)
- 6) Objectives of the project: Anticipated outcome, Communication with EFSA, Details of tasks to be fulfilled by each partner, Agreeing on timeframe (presentation from ALTERNATIVA leader)
- 7) Exchange of ideas on Final Report (including its publication as External Scientific Report or Event Report)
- 8) Planning of next meetings (teleconferences, other meetings, final meeting)

The final meeting occurred on 24<sup>th</sup> of June 2023, in hybrid format (on-line and EFSA), and considered the following topics for the agenda:

- 1) Welcome and opening
- 2) Adoption of the Agenda
- 3) Objectives of the project
- 4) General overview of the project activities and main outcomes (presentation from ALTERNATIVA leader)
- 5) Updates from each WP (presentation from ALTERNATIVA WPs leaders)
- 6) Feedback and discussion on the final report draft
- 7) Financial statement of the project – next steps
- 8) Future perspectives and project sustainability (presentation from ALTERNATIVA leader).

### 3.2 Action plan

Under the context of WP1, an action plan was developed, corresponding to the Deliverable 1 (Annex 1). The action plan established the timeline of all the planned activities, as well as their leader(s) and specific tasks. All actions were summarized and presented in chronological order. Additionally, the action plan document also summarized the interconnections between the different planned tasks.

### 3.3 Training activities

As described before, one of the goals of the ALTERNATIVA project was to enhance the capacity-building among members of the project team in applying methodologies related to the health RBA and sustainability impact assessment. Considering that, a training plan was organized, consisting in the Deliverable 2 (Annex 2).

Training activities were an opportunity to harmonize the knowledge between the different partners in i) risk-benefit assessment; and ii) sustainability assessment. All the participants were already experts in one domain and have acquired basic concepts to the second one. The topics covered are the followings and the detailed in the program (Annex 2):

- Health Food Risk-Benefit Assessment (3h00)
- Sustainability assessment (3h00)
- Impacts assessment in health and sustainability (2h00)
- Brainstorming session to generate ideas of strategies to integrate Health and sustainability Risk-Benefit Assessment into a common framework (2h30)

The completion of the trainings enabled to bridge these two domains, initiate the team and start to generate ideas on how to integrate health and sustainability assessment into a global approach, which served as the basis for the work developed under WP3, for the development of the protocol for the integration of risk-benefit assessment and sustainability (see section 3.4).

### 3.4 Protocol of integration of risk-benefit assessment and sustainability

Under the WP3, a protocol of integration of risk-benefit assessment and sustainability assessment was developed, comprising the Deliverable 3 (further in-depth information about the protocol will be provided in a scientific publication, highlighting the importance of integrating risk-benefit assessment with sustainability considerations). The protocol incorporates scientific and policy-relevant issues and is divided into three main parts (Figure 2):

- Part A – Initial Setting: Definition of the case study and indicators
- Part B – Implementation: Methodology options for data collection and assessment,
- Part C – Integration and Communication: Analysis and Communication.

#### 3.4.1 Part A – Initial Setting: Definition of the case study and indicators

The first step of the protocol consists of defining the case study in close collaboration with the policy and decision-makers, experts, and other stakeholders in the field. The definition of the case study must accomplish geographic and time scale boundaries, as well as precise definitions of alternative scenarios that are eventually to be studied.

The second step of the protocol is to establish a long list of indicators that are used for the sustainability and health risk-benefit assessment as measures of performance against sub-themes objectives. Indicators can be defined as “outcome variable” to be assessed under each domain (e.g., under the health domain: the disability-adjusted life years – DALYs) or sustainability theme (e.g., for measuring competitiveness and profitability). For each of the themes, indicators have been selected using several criteria, such as their significance in terms of assessing the relative domain, their relevance to the case study, their capacity to be measurable, their sensitiveness to variations, their capacity to be operationalized in a cost-effective way, etc. Following these criteria, we have defined a long list of indicators across several domains or themes defined in the four dimensions: health, economic, environment, and social. A description of the indicator is provided, as well as the reference to the document explaining how to calculate the indicator itself. This list can be adapted this list to add new indicators or modify the existing ones, following the criteria described above.

Then, the long list of indicators is downsized to a short list of core indicators that are most relevant for assessing sustainability and risk-benefits in the selected case study. The protocol does not provide a rigid short list of indicators, which necessarily will be also dependent on the case study peculiarity; instead, a set of general criteria or good practices derived from the literature are recommended for generating the short list of indicators:

- for each dimension (i.e. health, social, economic, and environmental), at least one indicator should be selected;
- the economic dimension should contain at least an indicator for competitiveness and profitability;
- preferably, broad indicators should be selected that provide a wide range of information;
- indicators should not be redundant, meaning that they should not “double-count” the same variable (e.g., morbidity, mortality and DALY);
- if a comparison between alternative products is foreseen, the indicator should be common between pairs and necessary for the comparison of performances of the products.

In general, data should be available to quantify each indicator identified in the short list. However, given the actual data availability, the final list of indicators is then defined. The final list of indicators should also be discussed with the policy and decision makers, and with experts and other influential stakeholders, to determine whether it contains all relevant indicators for assessing sustainability and risk-benefit of the case study.

### 3.4.2 Part B – Implementation: Methodology options for data collection and assessment

This part describes the process of gathering and measuring information on the indicators. Data collection is the process of gathering and measuring information on the indicators, in an established systematic fashion to evaluate outcomes (FAO, 2013b). Within an Integrated Health Risk-Benefit and Sustainability Assessment, primary and secondary data can be used. Primary data are original qualitative and quantitative data collected for a specific research

goal. Secondary data are derived from other sources such as literature or databases. The protocol provided recommendations for data collection based on previous experiences in sustainability and risk-benefit assessments, such as the comparability of sources, prioritization, and metadata documentation.

Furthermore, the protocol is also intended to highlight the dynamic aspects of the food system by laying out the interactions and various feedback mechanisms among the four dimensions, economic, environmental, health, and social. Indeed, various interactions and feedback loops could take place among the food system components, connecting the outcomes of the assessment across the different dimensions (Dentoni et al., 2023).

Moreover, if secondary data are used it is very important to evaluate the quality of data closely. This requires that additional information be archived with the data itself. Such information should at least include the study purpose, operationalization, data collection details, entities being studied and sampling criteria, and any known biases.

### 3.4.3 Part C – Integration: Analysis and Communication

The protocol aimed to provide a method to assess the health and sustainability of alternative food products, by establishing scenario comparison, multi-criteria analysis and/or graphical representation of the results. In multi-criteria analysis there are different ways to combine and compare different indicators across several dimensions.

One possibility is to combine them into a single composite indicator (Joint Research Centre-European Commission, 2008). The aggregation of diverse indicators in a single composite indicator differs conceptually from other indicator-based techniques such as Multi-Criteria Decision Assessments (MCDA) where the constituent indicators are not aggregated into a single number. Using MCDA it is also possible to develop scenario analysis and compare different scenarios (van Asselt et al., 2014). Scenario analysis is also very popular in health risk-benefit assessment (EFSA Scientific Committee, 2010). Here, the reference (or baseline) scenario often corresponds to the current or zero exposure to a dietary element, and the alternative scenarios concern the hypothetical consumers' exposure to the element under investigation (Boué et al., 2022; Naska et al., 2022). A similar dynamic methodology could be applied to estimate the social, economic and environmental impact of adopting alternative dietary behaviour under different scenarios. In this case, more complex econometric models are required, such as general equilibrium models (Mason-D'Croz et al., 2022), or behavioural models (Irz et al., 2019). Finally, in several cases, the use of radar charts or similar display formats has been applied to visualize the performances across different dimensions (Arfini & Bellassen, 2019).

All these methodologies have both pros and cons regarding the objective of the research that will be discussed. To identify the most appropriate approach to assess the health and sustainability of alternative food products, we referred to the literature on composite indicators and multi-criteria decision analysis, which is well established e.g. (Joint Research Centre-European Commission, 2008; Rogge, 2018; Zhou et al., 2012).

The protocol also includes the evaluation of uncertainties that propagate for the input factors to final values, with a positive or negative influence. Uncertainty analysis should be performed following approaches recommended by EFSA for case-specific assessments. The uncertainties in the body of evidence should be identified, including data gaps and needs for specific dimensions and themes, factors affecting the consistency of the results, the precision of the estimates, the internal (i.e. context-specific) and external validity (i.e. generalisability and replicability of the results) of the case studies.

The identification of data and knowledge gaps and needs is a specific output of the Integrated Risk-Benefit and Sustainability Assessment procedure.

As an additional output of the WP3, a scientific manuscript that will be submitted to a peer review international journal was drafted (this publication will provide further in-depth information about the protocol, highlighting the importance of integrating risk-benefit assessment with sustainability considerations).

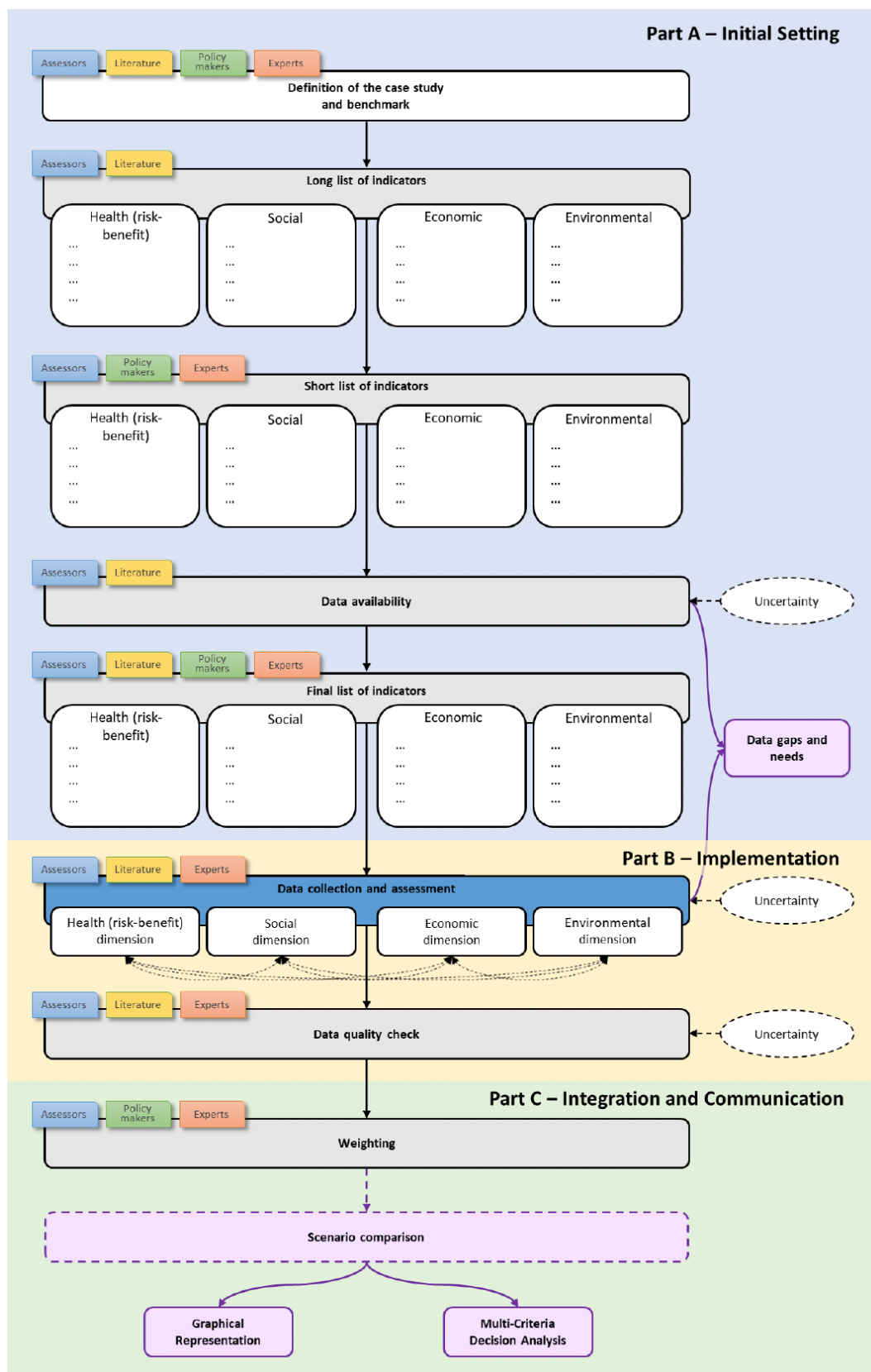


Figure 2. Protocol for an Integrated Risk-Benefit and Sustainability Assessment of Foods.

## 3.5 Case study: replacing red meat consumption by pulses in two countries, Portugal and Denmark

Under the WP4, an innovative model to assess the integrated impacts of different scenarios of food substitutions of red meat by pulses, in two different countries, Portugal and Denmark, were performed (Deliverable 5).

The main outputs obtained are summarized here.

A more detailed information about the case study is planned (as further detailed under the Communication plan, 3.7 below) in an international scientific publication, emphasizing the methodologies employed and the results obtained.

### 3.5.1 Social impact assessment

The substitution scenarios in both countries are shown in Table 2 and illustrate substantial positive impacts for all the aspects estimated: an increase of the AWU for all the categories considered (managers and employees) and a greater affordability.

Table 2. Social indicators for Denmark and Portugal in the reference and alternative scenarios.

| Social capital                                      | DENMARK |      |      |      |      | PORTUGAL |       |       |       |       |
|-----------------------------------------------------|---------|------|------|------|------|----------|-------|-------|-------|-------|
|                                                     | Current | AS1  | AS2  | AS3  | AS4  | Current  | AS1   | AS2   | AS3   | AS4   |
| Female manager/holder [AWU]                         | 520     | 882  | 1243 | 1605 | 1966 | 9140     | 9436  | 9732  | 10027 | 10323 |
| Farm labour force Female [AWU]                      | 1570    | 2678 | 3786 | 4894 | 6002 | 11110    | 11353 | 11596 | 11839 | 12082 |
| Female Farm managers, excluding group holding [AWU] | 280     | 456  | 632  | 807  | 983  | 3950     | 4236  | 4521  | 4807  | 5092  |
| Affordability [PPP]                                 | 1,10    | 0,85 | 0,61 | 0,36 | 0,12 | 0,38     | 0,31  | 0,24  | 0,17  | 0,10  |

The improvement is especially detectable in Denmark, where the total AWU will greatly increase in the alternative scenarios, as shown in Figure 3.

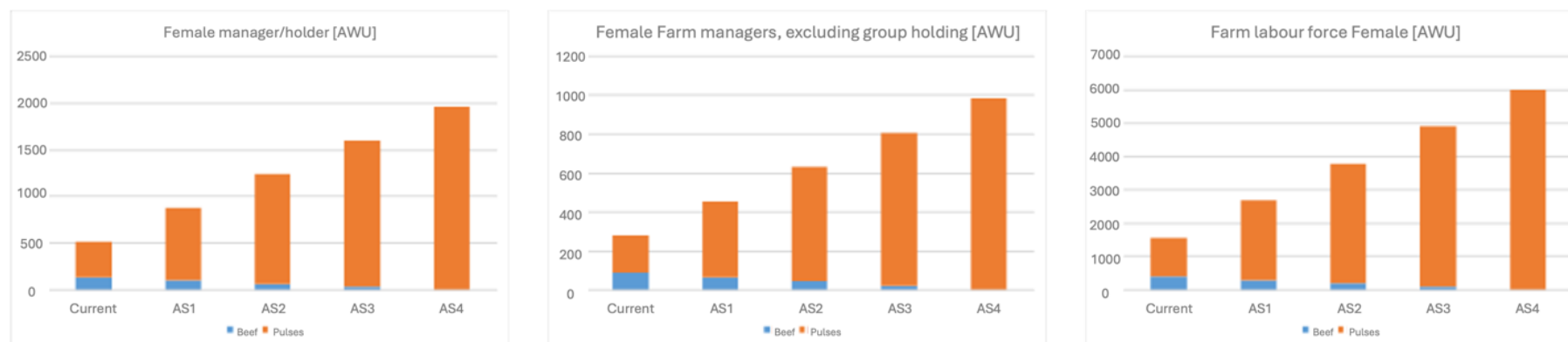


Figure 3. Female Labour Market in Denmark in the alternative scenarios.

In Portugal we estimated an improvement as well, even if not so significant as in the other case study (Figure 4).

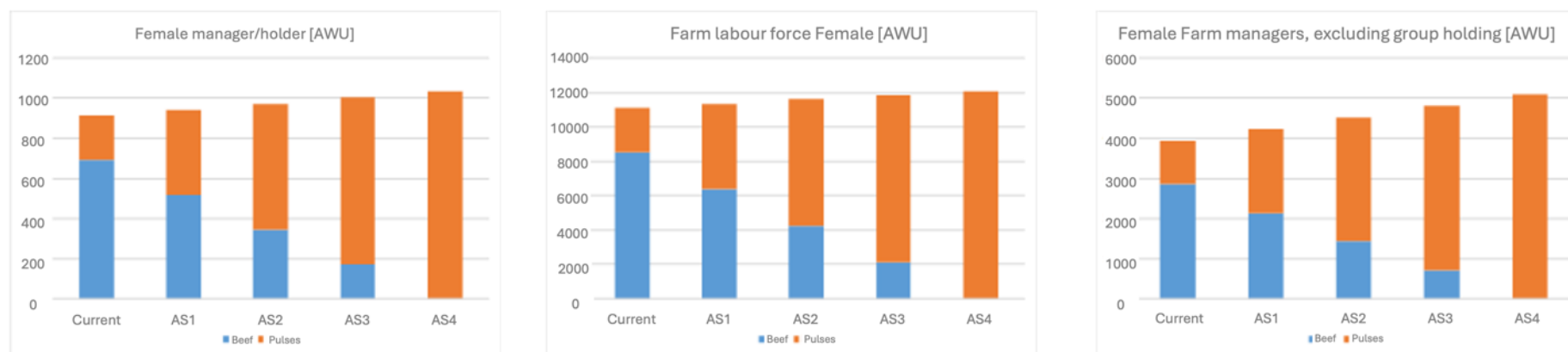


Figure 4. Female Labour Market in Portugal in the alternative scenarios.

The other dimension hugely affected in the estimated scenarios is the affordability, thanks to an increase of a cheaper protein source substituting a more expensive one (and, for Denmark, the gap is even wider), as we can observe in the graphs illustrating a drastic cost decrease in Figure 5 and Figure 6, respectively, in Portugal and Denmark.

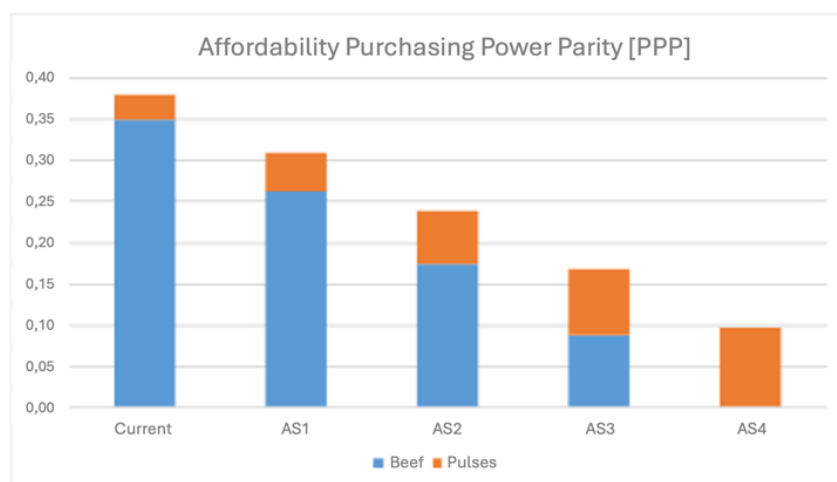


Figure 5. Affordability Purchasing Power Parity (PPP): Portugal.

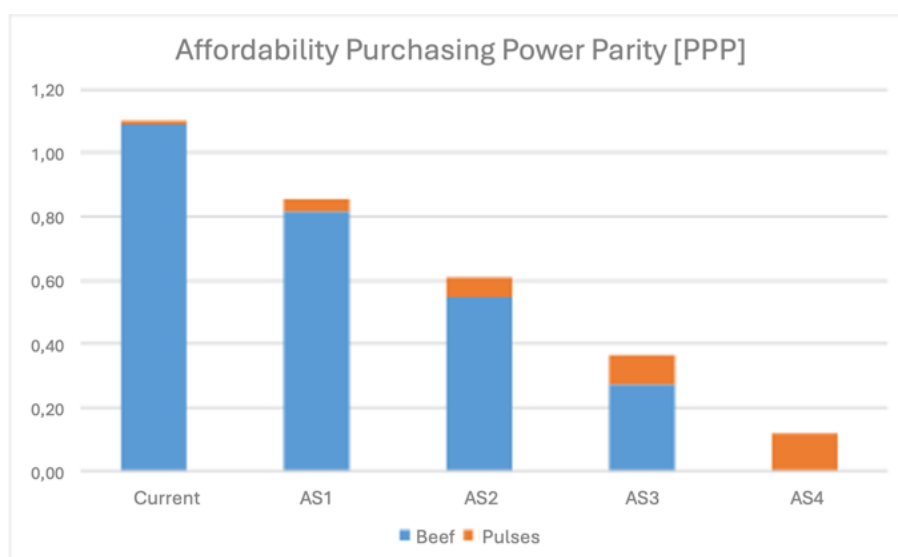


Figure 6. Affordability Purchasing Power Parity (PPP): Denmark.

### 3.5.2 Economic impact assessment

The results of the economic assessment for Denmark are reported in Figure 7. The gross operating margin (GOM), calculated as €/ha and as a % of turnover at farm level, is higher for beef production than for pulses production. This was reported by the secondary data both in €/ha and when assessed as % of turnover. In this latter case, the GOM is still lower for pulses than beef (median 32.6% lower). Interestingly, the net result, as % of turnover, is 50% higher for pulses than for beef. Apart from the likely incongruences of considering different sources from different countries, the lower net result for beef can be justified by the lower amount of fixed costs needed as compared to beef production. The performances of international trade indicators are consistently higher for beef than pulses, apart from the share of exports in non-EU countries. The price volatility for bovine carcass prices is 30% lower than for pulses. Overall, beef production seems more resilient and stable than pulses one. Considering regional workforce (NUTS2), all indicators show a significantly higher incidence of labour required in the case of beef production. Only considering the share of expenditure retained in the local economy (local multiplier), pulses are expected to perform equally or slightly better than animal production (+1.1%).

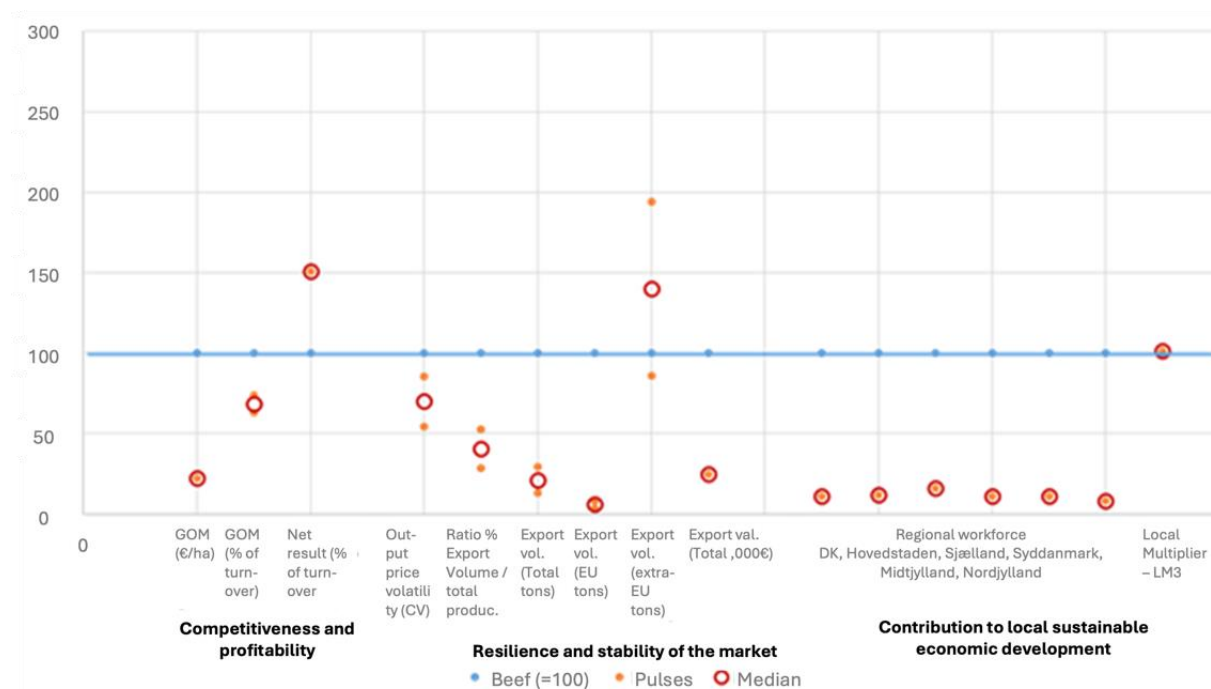


Figure 7. Difference (%) in economic indicators between pulses and beef in Denmark (beef = 100).

The results of the economic assessment for Portugal, as reported in Figure 8, are more heterogeneous than for Denmark. In Portugal, the financial performances at farm level seem higher for pulses than for beef. The gross operating margin (GOM) is around 50% higher for pulses production both in €/ha and as % of turnover. Net result is 7.6% higher for pulses (10.2% of turnover) than for beef (9.5%). The performances of international trade indicators

are consistently higher for beef than pulses, apart from the share of exports over total production. It can be presumed that some shares of imported pulses are also exported after being processed (e.g., processed and/or packaged). In Portugal, the price volatility indicator performs better for pulses than for beef; in other words, the results show that bovine carcass prices are 81% more volatile than pulses ones (CV= 17.8% vs. 9.8%). Finally, contribution to workforce is significantly higher for beef production, whilst the local multiplier for vegetables is performing equally or slightly better than for animal production (+1.1%).

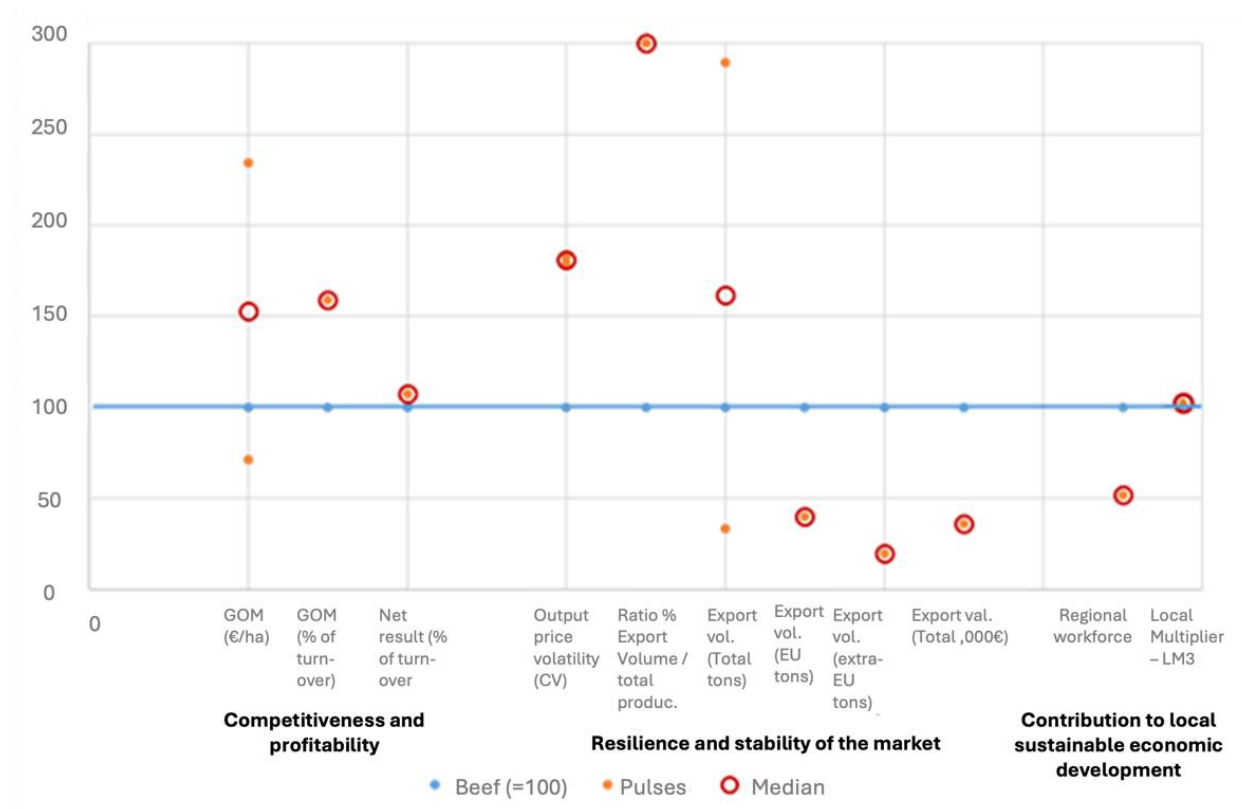


Figure 8. Portugal – Sustainability assessment: Economic dimension (each indicator is expressed as the % difference between pulses and beef = 100).

### 3.5.3 Environmental impact assessment

The results of the environmental assessment are reported in Figure 9. For Denmark, the major differences in absolute values that can be observed in the average environmental performances of the two food groups relate to GHG emissions (1.3 kg CO<sub>2</sub>eq per kg of pulses; 27.3 kg CO<sub>2</sub>eq per kg of beef), eutrophication (9.4 g PO<sub>4</sub><sup>3-</sup> eq per kg of pulses; 88.8 g PO<sub>4</sub><sup>3-</sup> eq per kg of beef), as well as biodiversity impact (135 \* 10<sup>14</sup> species.year for pulses; 1206.8 \* 10<sup>14</sup> species.year for beef). The latter indicator is strictly linked to the land use that is in turn constituted of two components, such as arable and pastureland use. The former is more than double for beef (16.6 m<sup>2</sup> per kg) compared to pulses (7.8 m<sup>2</sup> per kg); the latter is estimated to be 5.7 m<sup>2</sup> per kg of beef, while it is obviously null in the case of pulses.

Considering water use, the consumption of pulses impacts on average half compared to the beef one (358.6 L per kg of pulses, 719.6 per kg of beef).

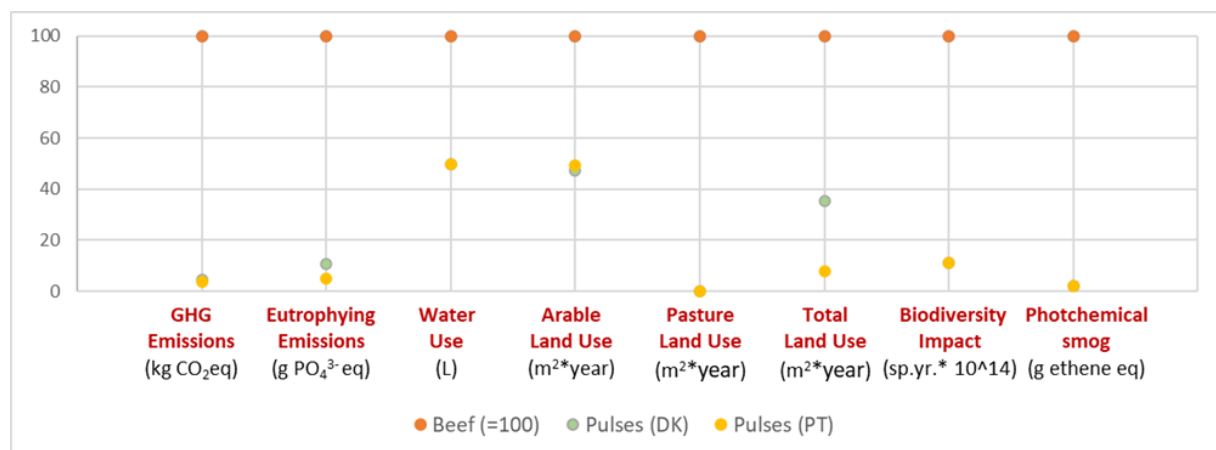


Figure 9. Environmental dimension (each indicator is expressed as the % difference between pulses and beef, where beef is equal to 100) for Denmark and Portugal. The reference values used for the estimation of photochemical smog equal those applied to Denmark as no specific data are available for Portugal.

Overall, the results of the environmental assessment for Portugal are very similar compared to those indicated for Denmark. The contribution to environmental degradation and resource use was larger for beef, with GHG emissions, eutrophying emissions, and land use showing the greatest differences between the two food groups. Per kg of beef, GHG emission, eutrophication and total land use are respectively 27 times (34.1 kg CO<sub>2</sub>eq), 21 times (194.9 g PO<sub>4</sub><sup>3-</sup>eq) and 13 times (98.1 m<sup>2</sup>) the values associated to pulses (1.3 kg CO<sub>2</sub>eq, 9.4 g PO<sub>4</sub><sup>3-</sup>eq, and 7.8 m<sup>2</sup>, respectively). Arable land use and water use are around double for beef (15.9 m<sup>2</sup> and 718.6 L, respectively) compared to pulses (7.8 m<sup>2</sup> and 357.6 L, respectively).

The gap between the land use required for pulses and beef production seemed to be lower in Portugal compared to Denmark. This difference can be explained by a wider pastureland use for kg of beef in Portugal. The arable land use per kg of the two food groups is relatively similar in the two countries, however, the freshwater and marine eutrophication potential is higher in Portugal. In addition, the values referred to global biodiversity impact and photochemical smog for the two food groups are the same when comparing Denmark and Portugal, due to the difficulty in referring the related assessments at the country level.

The estimated environmental impacts associated to current and alternative consumption of pulses and beef calculated for Denmark and Portugal are shown in Table 3. As expected, all the environmental indicators are estimated to decrease moving from the reference to the alternative scenarios, with the best performance referred to the situation in which beef consumption is totally replaced by pulses (AS4). In this scenario the highest percentages of change (above 88%) compared to the reference situation can be observed for GHG, eutrophying emissions, photochemical smog, as well as for pastureland use, in both case studies. For the same scenario comparison, the change in total land use remains high in

www.efsa.europa.eu/publications

Portugal (88.8%), while decrease in Denmark (62.8%). On the other hand, changes in arable land use and water land use are less relevant being around 40% in Portugal and around 50% in Denmark (data not shown).

Table 3. Environmental impacts associated to current and alternative consumption scenarios for beef and pulses in Denmark (DK) and Portugal (PT).

| Environmental indicators                               | Ref. |      | AS1  |      | AS2  |      | AS3  |      | AS4  |      |
|--------------------------------------------------------|------|------|------|------|------|------|------|------|------|------|
|                                                        | DK   | PT   | DK   | PT   | DK   | PT   | DK   | PT   | DK   | PT   |
| GHG emissions (kg CO <sub>2</sub> eq)                  | 1.2  | 0.8  | 0.9  | 0.6  | 0.6  | 0.4  | 0.4  | 0.2  | 0.1  | 0.0  |
| Eutrophication (g PO <sub>43-</sub> eq)                | 4.0  | 4.5  | 3.1  | 3.5  | 2.2  | 2.4  | 1.3  | 1.4  | 0.5  | 0.3  |
| Water Use (L)                                          | 33.5 | 19.9 | 29.5 | 17.8 | 25.5 | 15.8 | 21.4 | 13.7 | 17.4 | 11.7 |
| Arable Land Use (m <sup>2</sup> *year)                 | 0.8  | 0.4  | 0.7  | 0.4  | 0.6  | 0.3  | 0.5  | 0.3  | 0.4  | 0.3  |
| Pasture Land Use (m <sup>2</sup> *year)                | 0.3  | 1.9  | 0.2  | 1.4  | 0.1  | 0.9  | 0.1  | 0.5  | 0.0  | 0.0  |
| Total Land Use (m <sup>2</sup> *year)                  | 1.0  | 2.3  | 0.9  | 1.8  | 0.7  | 1.3  | 0.5  | 0.8  | 0.4  | 0.3  |
| Biodiversity impact (species.year * 10 <sup>14</sup> ) | 54.4 | 28.6 | 42.4 | 22.6 | 30.5 | 16.5 | 18.5 | 10.5 | 6.6  | 4.4  |
| Photochemical smog (g ethene eq)                       | 0.3  | 0.1  | 0.2  | 0.1  | 0.1  | 0.1  | 0.1  | 0.0  | 0.0  | 0.0  |

**AS: Alternative scenario. AS1: substitution of 25% of beef consumption by pulses. AS2: substitution of 50% of beef consumption by pulses. AS3: substitution of 75% of beef consumption by pulses. AS4: substitution of 100% of beef consumption by pulses.**

### 3.5.4 Health impact assessment

The current consumption of unprocessed beef and pulses in the Danish and Portuguese populations varies as presented in Table 4. It was observed that consumption of beef is higher in Denmark (mean 44.6 grams per day, standard deviation (sd) 38,8 g/day), and of pulses is higher in Portugal (mean 10.1 g/day, sd 21.6). Because an estimation of the impact of different possibilities of food substitution was performed, these differences are also translated in the alternative scenarios (AS).

Table 4. Consumption of unprocessed beef and pulses (g/day) in the Portuguese and Danish populations in the reference and alternative scenarios.

| Scenario  | Denmark |      |       |        |      |      |       |        |
|-----------|---------|------|-------|--------|------|------|-------|--------|
|           | Pulses  |      |       |        | Beef |      |       |        |
|           | Mean    | sd   | 2.50% | 97.50% | Mean | sd   | 2.50% | 97.50% |
| Reference | 4.1     | 4.9  | 0     | 17.5   | 44.6 | 38.8 | 0.9   | 149    |
| AS1       | 15.3    | 11.1 | 1.2   | 43.4   | 33.4 | 29.1 | 0.7   | 111.8  |
| AS2       | 26.4    | 20.3 | 1.8   | 79     | 22.3 | 19.4 | 0.5   | 74.5   |

|                  |             |           |              |               |             |           |              |               |
|------------------|-------------|-----------|--------------|---------------|-------------|-----------|--------------|---------------|
| <b>AS3</b>       | 37.6        | 29.8      | 2.4          | 115.7         | 11.1        | 9.7       | 0.2          | 37.3          |
| <b>AS4</b>       | 48.7        | 39.4      | 2.9          | 153.1         | 0           | 0         | 0            | 0             |
| <b>Portugal</b>  |             |           |              |               |             |           |              |               |
| <b>Pulses</b>    |             |           |              |               | <b>Beef</b> |           |              |               |
|                  | <b>Mean</b> | <b>sd</b> | <b>2.50%</b> | <b>97.50%</b> | <b>Mean</b> | <b>sd</b> | <b>2.50%</b> | <b>97.50%</b> |
| <b>Reference</b> | <b>10.1</b> | 21.6      | 0            | 71.9          | <b>22.6</b> | 56        | 0            | 194.9         |
| <b>AS1</b>       | 15.8        | 26        | 0            | 85.7          | 16.9        | 42        | 0            | 146.2         |
| <b>AS2</b>       | 21.4        | 35.8      | 0            | 121           | 11.3        | 28        | 0            | 97.5          |
| <b>AS3</b>       | 27.1        | 47.7      | 0            | 162.9         | 5.6         | 14        | 0            | 48.7          |
| <b>AS4</b>       | 32.71705    | 60.5      | 0            | 201.1         | 0           | 0         | 0            | 0             |

AS: Alternative scenario. AS1: substitution of 25% of beef consumption by pulses. AS2: substitution of 50% of beef consumption by pulses. AS3: substitution of 75% of beef consumption by pulses. AS4: substitution of 100% of beef consumption by pulses. Sd: Standard deviation. 2.5% and 97.5%: percentiles.

A positive health impact of all scenarios of substitution in the two populations was observed. Results per 100.000 inhabitants were presented, to allow for comparisons between countries. The net-DALY gain (which reflects the impact of reducing the risk of CRC, diabetes and IHD as consequence of reducing beef and increasing pulses consumption) was higher for each increase in substitution (Table 5). The estimated positive impact was more substantial for the Danish population in all scenarios.

Table 5. Estimated difference in years of life lived with disability (YLD), years of life lost (YLL) and disability-adjusted life years (DALYs) per 100.000 between reference and alternative scenario (AS1-AS4) in Portugal and Denmark (mean and 95% uncertainty interval (UI)).

|     |                | YLD  |        |       | YLL   |        |        | DALY  |        |        |
|-----|----------------|------|--------|-------|-------|--------|--------|-------|--------|--------|
|     | Health outcome | Mean | 95% UI |       | Mean  | 95% UI |        | Mean  | 95% UI |        |
|     | Portugal       |      |        |       |       |        |        |       |        |        |
| AS4 | CRC            | -0.3 | 0.0    | -3.3  | -5.7  | 0.0    | -54.9  | -6.0  | 0.0    | -58.2  |
|     | Diabetes       | -1.0 | 0.0    | -8.7  | -0.4  | 0.0    | -2.7   | -1.4  | 0.0    | -11.2  |
|     | IHD            | -0.6 | 0.0    | -5.3  | -12.0 | 0.0    | -111.3 | -12.6 | 0.0    | -116.2 |
|     | TOTAL          | -1.9 | 0.0    | -17.3 | -18.1 | 0.0    | -168.8 | -20.0 | 0.0    | -185.6 |
|     |                |      |        |       |       |        |        |       |        |        |
| AS3 | CRC            | -0.3 | 0.0    | -2.5  | -4.4  | 0.0    | -42.3  | -4.6  | 0.0    | -44.8  |
|     | Diabetes       | -0.8 | 0.0    | -6.8  | -0.3  | 0.0    | -2.1   | -1.1  | 0.0    | -8.8   |
|     | IHD            | -0.4 | 0.0    | -4.2  | -9.6  | 0.0    | -89.8  | -10.0 | 0.0    | -93.8  |
|     | TOTAL          | -1.5 | 0.0    | -13.6 | -14.3 | 0.0    | -134.2 | -15.8 | 0.0    | -147.4 |
|     |                |      |        |       |       |        |        |       |        |        |
| AS2 | CRC            | -0.2 | 0.0    | -1.7  | -3.0  | 0.0    | -29.0  | -3.2  | 0.0    | -30.7  |
|     | Diabetes       | -0.5 | 0.0    | -4.7  | -0.2  | 0.0    | -1.4   | -0.7  | 0.0    | -6.1   |
|     | IHD            | -0.3 | 0.0    | -3.0  | -6.8  | 0.0    | -64.6  | -7.2  | 0.0    | -67.5  |
|     | TOTAL          | -1.0 | 0.0    | -9.5  | -10.0 | 0.0    | -95.0  | -11.1 | 0.0    | -104.3 |

|            |                 | YLD  |        |       | YLL   |        |        | DALY  |        |        |
|------------|-----------------|------|--------|-------|-------|--------|--------|-------|--------|--------|
|            | Health outcome  | Mean | 95% UI |       | Mean  | 95% UI |        | Mean  | 95% UI |        |
|            |                 |      |        |       |       |        |        |       |        |        |
| <b>AS1</b> | <b>CRC</b>      | -0.1 | 0.0    | -0.9  | -1.5  | 0.0    | -14.9  | -1.6  | 0.0    | -15.8  |
|            | <b>Diabetes</b> | -0.3 | 0.0    | -2.5  | -0.1  | 0.0    | -0.7   | -0.4  | 0.0    | -3.2   |
|            | <b>IHD</b>      | -0.2 | 0.0    | -1.6  | -3.7  | 0.0    | -35.0  | -3.8  | 0.0    | -36.5  |
|            | <b>TOTAL</b>    | -0.5 | 0.0    | -5.0  | -5.3  | 0.0    | -50.6  | -5.9  | 0.0    | -55.5  |
|            | <b>Denmark</b>  |      |        |       |       |        |        |       |        |        |
| <b>AS4</b> | <b>CRC</b>      | -1.1 | 0.0    | -4.7  | -19.8 | 0.0    | -84.8  | -20.9 | 0.0    | -89.2  |
|            | <b>Diabetes</b> | -2.6 | 0.0    | -11.2 | -1.5  | 0.0    | -9.7   | -4.1  | 0.0    | -20.1  |
|            | <b>IHD</b>      | -0.8 | 0.0    | -3.0  | -13.9 | 0.0    | -54.3  | -14.7 | 0.0    | -56.8  |
|            | <b>TOTAL</b>    | -4.5 | 0.0    | -18.9 | -35.3 | 0.0    | -148.8 | -39.7 | 0.0    | -166.2 |
|            |                 |      |        |       |       |        |        |       |        |        |
| <b>AS3</b> | <b>CRC</b>      | -0.8 | 0.0    | -3.6  | -15.1 | 0.0    | -65.0  | -15.9 | 0.0    | -68.4  |
|            | <b>Diabetes</b> | -2.0 | 0.0    | -8.6  | -1.2  | 0.0    | -7.4   | -3.1  | 0.0    | -15.5  |
|            | <b>IHD</b>      | -0.6 | 0.0    | -2.4  | -10.9 | 0.0    | -43.4  | -11.5 | 0.0    | -45.5  |
|            | <b>TOTAL</b>    | -3.4 | 0.0    | -14.7 | -27.1 | 0.0    | -115.9 | -30.5 | 0.0    | -129.4 |
|            |                 |      |        |       |       |        |        |       |        |        |
| <b>AS2</b> | <b>CRC</b>      | -0.6 | 0.0    | -2.5  | -10.2 | 0.0    | -44.3  | -10.7 | 0.0    | -46.7  |
|            | <b>Diabetes</b> | -1.3 | 0.0    | -5.9  | -0.8  | 0.0    | -5.1   | -2.1  | 0.0    | -10.6  |
|            | <b>IHD</b>      | -0.4 | 0.0    | -1.7  | -7.5  | 0.0    | -30.9  | -7.9  | 0.0    | -32.4  |
|            | <b>TOTAL</b>    | -2.3 | 0.0    | -10.1 | -18.5 | 0.0    | -80.3  | -20.8 | 0.0    | -89.6  |
|            |                 |      |        |       |       |        |        |       |        |        |
| <b>AS1</b> | <b>CRC</b>      | -0.3 | 0.0    | -1.3  | -5.2  | 0.0    | -22.7  | -5.4  | 0.0    | -23.9  |
|            | <b>Diabetes</b> | -0.7 | 0.0    | -3.1  | -0.4  | 0.0    | -2.6   | -1.1  | 0.0    | -5.4   |
|            | <b>IHD</b>      | -0.2 | 0.0    | -0.9  | -3.9  | 0.0    | -16.6  | -4.1  | 0.0    | -17.3  |
|            | <b>TOTAL</b>    | -1.2 | 0.0    | -5.2  | -9.5  | 0.0    | -41.8  | -10.7 | 0.0    | -46.6  |

Our results show that the reduction of the risk of CRC was the largest contribution for the positive health effect in Denmark in all scenarios, whereas prevention of IHD had the largest impact in Portugal (Figure 10).

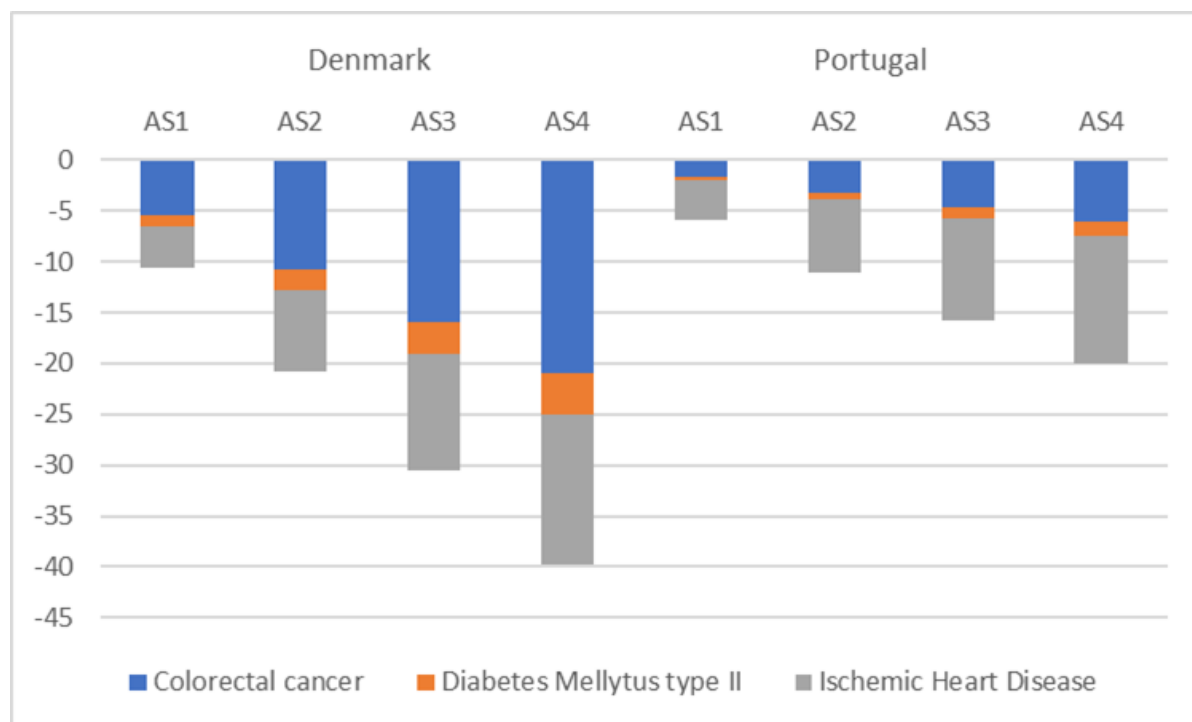


Figure 10. Difference DALYs per 100.000 between reference and alternative scenario (AS1-AS4) in Portugal and Denmark.

### 3.5.5 Multi-criteria decision analysis

Under the WP4, a MCDA approach was also considered and implemented.

A total of 11 experts were involved in the MCDA model development process. Most of the experts were part of the ALTERNATIVA consortium. The exceptions were one expert that contributed to identifying criteria with the environmental dimension, and one expert with animal welfare, which was missing from the consortium.

The number of criteria initially defined within each dimension varied. For example, “diet-related impacts” was the only criteria defined within “health”, whereas four criteria were identified within the “social” dimension. After the reflection period, and throughout the criteria operationalization meetings, the value tree was refined, until the current set of 15 criteria was settled (Figure 11). The criteria can be described as follows:

- **Profitability across the value chain:** Considers the profitability of all production processes across the supply chain.
- **Affordability for consumers:** Considers how affordable the product is for the final consumer.
- **Local economic development:** Considers the impacts to local economies, based on employment and economic spill-over on the territory.
- **Impact on local communities:** Considers how much traditional practices and historical heritage are impacted by the change.

- **Acceptance:** Considers how much people are willing to change and if there are food related misconceptions that may have an impact.
- **Fair/Ethical practices:** Whether the production process considers the respect for fair and ethical practices, namely regarding gender equity.
- **Accessibility:** Considers the ease of access to the foods.
- **Animal Welfare:** Considers whether the production process follows animal welfare good practices.
- **Diet and Food Related Health Impacts:** Considers how the change can impact people's health, be it through diet related disease, food safety hazards, or nutritional status.
- **Environment Related Health Impacts:** Considers how the change can impact people's health, through the environmental impacts of the production process.
- **Biodiversity:** Considers the food products' impact on biodiversity.
- **Climate Change:** Considers the food products' production process contribution to climate change.
- **Water Use:** Considers the food products' impact on the use of water resources.
- **Land Use:** Considers the food products' impact on the use of land resources (arable and pasture).
- **Pollution:** Considers the pollution caused by the production process.

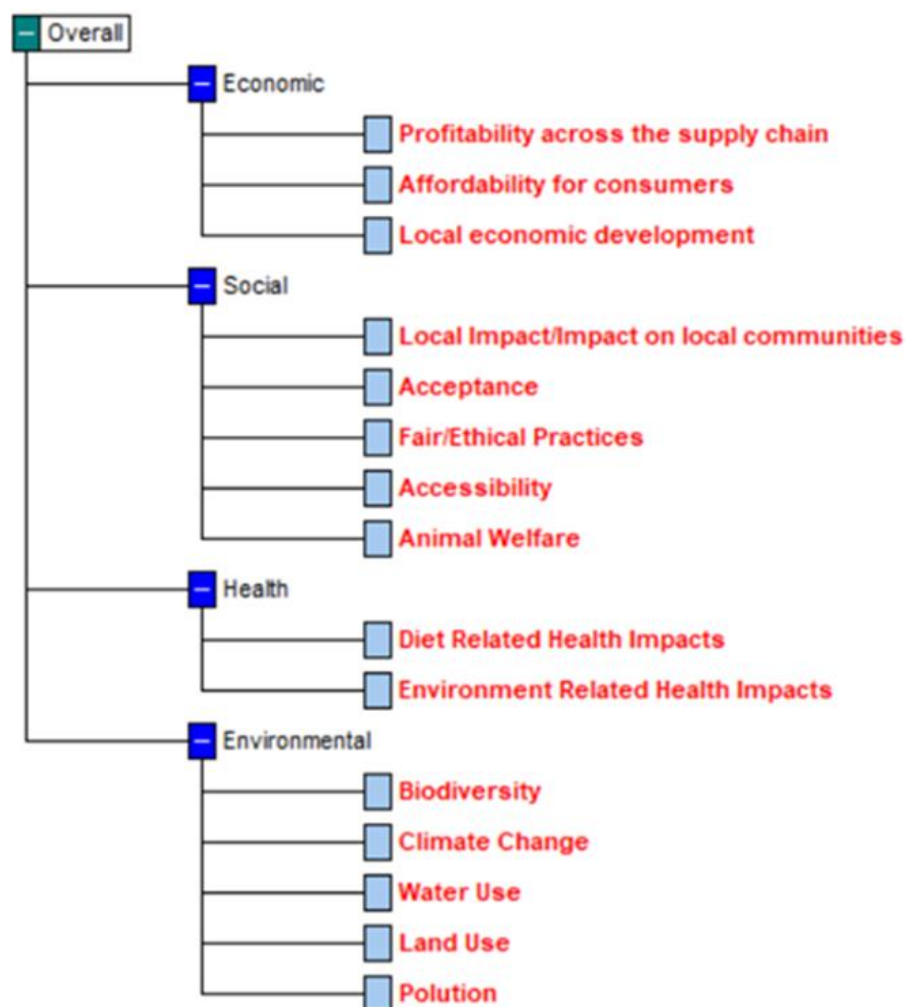


Figure 11. Final value tree of the multi-criteria decision analysis model for estimating impacts of dietary substitutions.

When the results of 2.4.1. to 2.4.4. were presented to the experts and integrated in the final evaluation process, the MCDA model showed that the most attractive scenario was the 100% substitution scenario (Figure 12, example of Danish case study). This was in line with the expectations of the experts and no doubts arose at this point.

Table of scores

| Options       | Overall | Profit | Affordability | LED    | Local  | Accept  | F/E prac | N16    | DALY   | ERHI   | Biodiv. | Climate | WU     | LU     | Polution |
|---------------|---------|--------|---------------|--------|--------|---------|----------|--------|--------|--------|---------|---------|--------|--------|----------|
| [ all upper ] | 100.00  | 100.00 | 100.00        | 100.00 | 100.00 | 100.00  | 100.00   | 100.00 | 100.00 | 100.00 | 100.00  | 100.00  | 100.00 | 100.00 | 100.00   |
| 100%          | 91.89   | -83.83 | 396.67        | -66.67 | 0.00   | -233.33 | 127.27   | 0.00   | 39.70  | 145.00 | 61.34   | 122.22  | 88.90  | 173.33 | 125.00   |
| 75%           | 67.82   | -60.79 | 323.33        | -66.67 | 0.00   | -233.33 | 127.27   | 0.00   | 30.50  | 115.00 | 24.36   | 66.67   | 82.90  | 166.67 | 92.50    |
| 50%           | 43.02   | -39.50 | 246.67        | -66.67 | 100.00 | -233.33 | 127.27   | 0.00   | 20.80  | 80.00  | -9.47   | 0.00    | 76.75  | 153.33 | 70.00    |
| 25%           | 28.67   | -19.75 | 160.00        | -66.67 | 100.00 | 0.00    | 127.27   | 0.00   | 10.70  | 45.00  | -54.54  | -55.56  | 70.75  | 140.00 | 47.50    |
| 0%            | 2.27    | 0.00   | 0.00          | 66.67  | 0.00   | 100.00  | 0.00     | 0.00   | 0.00   | 10.00  | -93.60  | -111.11 | 64.75  | 133.33 | 25.00    |
| [ all lower ] | 0.00    | 0.00   | 0.00          | 0.00   | 0.00   | 0.00    | 0.00     | 0.00   | 0.00   | 0.00   | 0.00    | 0.00    | 0.00   | 0.00   | 0.00     |
| Weights :     |         | 0.0605 | 0.0968        | 0.0403 | 0.0081 | 0.0524  | 0.0282   | 0.0202 | 0.1210 | 0.0766 | 0.1129  | 0.1250  | 0.1048 | 0.0685 | 0.0847   |

Figure 12. Table of scores for the Danish case study.

### 3.6 Stakeholders' current understanding and attitudes towards a risk-benefit assessment on health and sustainability aspects

Under WP5, activities and tasks were developed to characterize the stakeholders' current understanding and attitudes towards a risk-benefit assessment on health and sustainability aspects. The obtained outputs were summarized in Deliverable 6 (further detailed information on the questionnaire development process will be provided in a scientific publication, highlighting the key milestones and achievements).

In the first round, risk assessors, risk managers and communicators exchanged ideas around relevant topics that could be included in the questionnaires to address stakeholders' current understanding, attitudes, and perceptions towards a risk-benefit assessment on health and sustainability. Following views exchanged and proposals made during the first round of discussions, draft-preliminary versions of RBA-related questionnaires have been prepared.

#### 3.6.1 Main inputs of the first round of stakeholder consultations

- Risk Managers-Risk Communicators

Several risk managers and communicators participating in round 1 had almost no knowledge or prior experience in the application of an RBA methodology. To enhance and streamline the group's understanding, the RBA steps (e.g., definition of the referent and alternative scenarios, identification and selection of components and indicators related to health and sustainability aspects, characterization of health effects, exposure assessment, risk/benefit characterization, health and sustainability impact quantification, comparison of scenarios) were presented at the beginning of the session.

Through the discussion, most participants recognized the importance of weighing risks and benefits in the assessment of food-related matters. Nonetheless, those in the group who have been exposed to RBA-related projects expressed their concern as they felt that the implementation of certain steps (e.g., risk/benefit characterization, integrating health and sustainability) form part of risk management decisions. Within this context, participants indicated their reluctance towards a generalized application of RBA, suggesting that the weighing of risks and benefits, particularly when considering economic, cultural, and social factors should be addressed by risk managers and not risk assessors.

Participants further noted that RBA is not currently a topic of discussion within their organizations. It was, therefore, realized that the dimensions of knowledge and experience related to different aspects of RBA and sustainability should be captured through the questionnaire.

Among the aspects of RBA that have been considered relevant to be assessed through the developed questionnaire, are: (a) the definition of risk and benefit, (b) the country-specific differences in the perception of risk and (c) the expertise available in each Organization for

[www.efsa.europa.eu/publications](http://www.efsa.europa.eu/publications)

every RBA step. Participants acknowledged that the concept of risk and benefit and the balance in-between should be well established and clarified. They further highlighted the added value of a harmonized common approach to be applied in all EU MS. Participants further recognized that policy makers may need targeted education and training to implement and interpret RBA methods and results. Establishing networks that would include experts from different fields was identified as a necessity to apply, disseminate and communicate RBA findings. The importance of teamwork in the implementation of an RBA including sustainability aspects was also indicated.

Risk managers commented that the anticipated challenges and associated uncertainties in the application of this integrated framework are mainly related to the availability of data from different countries. To reach a conclusion, discrepancies in the perception of risk have to be reconciled. Participants further agreed that indices related to politics, economy, culture, legislative processes also have to be considered. Finally, it was proposed that the harmonization of methods could be achieved at the European level raising however the need for developing appropriate guidelines for method implementation.

Overall, risk managers and communicators participating in round 1 were not particularly aware of RBA steps and related methods, were questioning the definition of benefit and were reluctant to introduce RBA approaches in their work, although they acknowledged that the enhancement of an approach that would holistically address food safety/quality is needed.

- Risk Assessors

In contrast, participants with expertise in risk assessment were aware of the multi-dimensional RBA framework, including expertise from different fields such as nutrition, toxicology, microbiology, epidemiology and data modelling, that provides insight into the assessment of public health impact of food choices. Most risk assessors have been previously involved in RBA projects. Some of them were aware of the 2010 EFSA guidance document on RBA. However, it also became apparent that stakeholders felt reluctant about the potential of integrating sustainability aspects within an RBA framework focusing on health.

Experts also noted that it might be useful to know in advance how risk assessment and management are implemented in each country and how the integration of sustainability may affect the independency of risk assessment. They also pointed out that the relative importance of societal needs should be assessed well in advance. In addition, the problem formulation (e.g. at substance, food, or eating pattern level) should be clearly addressed in an integrated framework.

Similarly to risk managers, assessors also identified lack of or limited accessibility to data as potential challenges and source of uncertainty. It was noted that the introduction of additional parameters (sustainability indicators; economic, cultural, social aspects) may increase the difficulties and it can intimidate assessors to engage in this work. Specifically, during the implementation of an integrated RBA framework, safeguarding the independence of risk assessment from risk management is important. Additionally, the need for a realistic timeframe should also be taken into consideration. Finally, the need of a harmonized approach to be used at national and EU level was also raised.

Overall, risk assessors were more familiar with RBA methods and results, but were reluctant to embark into the integration of health and sustainability aspects in RBA modelling.

### 3.6.2 Main inputs of the second round of stakeholder consultations

Based on the discussion points and remarks collected through round 1, it was decided that two questionnaires are to be developed; one to assess individual capacities within an organization's staff and one to be filled at the level of organization or any related corporation.

In the second round, participants were allocated to two new groups: (a) the first one included EFSA representatives and (b) the second included experts from national food authorities. During these sessions, participants discussed the content, proposed questions and responses included in the questionnaires. Among the main subjects/issues discussed were key concepts that should be well established and clarified for a risk assessor/manager/communicator when initiating and applying an RBA approach, familiarity with RBA methodology, perspectives about the most relevant context within which the questionnaire should be applied, challenges and uncertainties associated with this integrated framework (data availability being the most important).

## 3.7 Communication: main outputs, policy brief and web-assisted international meeting

Concerning the communication in the context of ALTERNATIVA project, three aspects should be considered: i) the outputs produced in the context of the project; ii) the policy brief; and, iii) the web-assisted international meeting.

- Outputs obtained in the context of ALTERNATIVA project

Several outputs were planned in project proposal. All of them were attained and some of them were surpassed, as detailed in Table 6.

Regarding the publications, two national (Annexes 4 and 5) and one international manuscripts were drafted, according to the expected outputs (Table 6). Concerning the international manuscript, the Consortium aims to submit it in a high-impact journal (e.g. Nature Food, Comprehensive Reviews in Food Science and Food Technology, Trends in Food Sciences & Technology) early in 2025.

Table 6. List of outputs planned and obtained under ALTERNATIVA project.

| <b>List of outputs</b>                                                | <b>2022</b> | <b>2023</b> | <b>Total planned</b> | <b>Total obtained</b> |
|-----------------------------------------------------------------------|-------------|-------------|----------------------|-----------------------|
| <b>Publications</b>                                                   |             |             |                      |                       |
| <i>Papers in international journals</i>                               |             | 1           | 1                    | 1                     |
| <i>Papers in national journals</i>                                    | 1           | 1           | 2                    | 2                     |
| <b>Communications</b>                                                 |             |             |                      |                       |
| <i>Communications in international meetings</i>                       | 1           | 2           | 3                    | 4                     |
| <i>Communications in national meetings</i>                            | 1           | 1           | 2                    | 3                     |
| <b>Case study report</b>                                              |             | 1           | 1                    | 1                     |
| <b>Final report</b>                                                   |             | 1           | 1                    | 1                     |
| <b>Organization of a web-assisted international meeting</b>           |             | 1           | 1                    | 1                     |
| <b>Protocol: RBA and Sustainability impact assessment integration</b> |             | 1           | 1                    | 1                     |
| <b>Policy brief</b>                                                   |             | 1           | 1                    | 1                     |

Concerning the communications, a total of seven communications were performed, i.e., three communications in national meetings (Noite Europeia dos investigadores; Webinar APTAC; 1<sup>st</sup> Egas Moniz One Health Symposium), and four communications in international meetings (ONE Conference; 1<sup>st</sup> International Conference on Food, Nutrition & Public Health; International Congress S4agro 2023; 6<sup>th</sup> International Congress of the Egas Moniz Center for Interdisciplinary Research), two of them as oral communications.

As introduced before, the protocol of integration of risk-benefit and sustainability assessment of food was developed (Figure 2). A case study report was produced (3.5 above, Deliverable 5), summarizing the major outputs from the case study of substituting red meat consumption by pulses, in Portugal and Denmark.

- Policy brief

A policy brief was developed to disseminate the new framework developed under ALTERNATIVA, i.e., the integration of health and sustainability assessments, thus presenting the approach used by the project team and the results obtained in each dimension for the current scenario of consumption of beef and pulses (Deliverable 7, Annex 3).

- Web-assisted international meeting

Under the context of ALTERNATIVA project, a web-assisted international meeting was organized, and occurred on 20<sup>th</sup> June 2023 (Annex 6). This workshop aimed to promote the development of critical mass thinking in the areas of expertise covered in the ALTERNATIVA project. During this meeting a plenary lecture by Prof Daniele del Rio took place as well as the main results and achievements obtained in each work package.

## 4 Conclusion

“ALTERNATIVA | Alternative protein sources in the European diets – integrating health risk-benefit and sustainability” aimed to develop a holistic approach to assess the impact of alternative protein sources integrating simultaneously risk-benefit for human health, and sustainability aspects (environment, economy and society). Combining expertise in risk-benefit assessment and sustainable impact assessment, ALTERNATIVA enhanced the capacity of involved consortium to apply this holistic perspective to the evaluation of the impact of replacing red meat consumption by pulses, in two different countries, Portugal and Denmark.

As planned, ALTERNATIVA expanded and progressed on dimensions usually addressed separately, i.e. health risk-benefit assessment and sustainability impact assessment, developing a harmonized framework that facilitates the integration of outputs and the communication of results to the involved stakeholders.

Moreover, the established framework was applied in the context of a case study dealing with the assessment of the impact of replacing red meat consumption by pulses in Portugal and Denmark. The health impact of different scenarios of replacement of red meat with pulses was measured in terms of “disability-adjusted life year” (DALY). The environmental, economic and social sustainability of red meat and pulses were also assessed using different sustainability indicators. MCDA was developed by involving experts from the different domains who met in multiple meetings to develop the value tree and define criteria within each dimension. A positive health impact of all scenarios of the dietary substitution on the two populations was estimated. The impact of reducing the risk of CRC, diabetes, and IHD because of reducing beef and increasing pulses consumption was higher for each increase in substitution. The estimated positive impact was larger for the Danish population than for the Portuguese, which is justifiable given the baseline consumption of red meat in both countries (higher in Denmark). It varied between 5.9 DALYs (25% substitution) and 20 DALYs per 100,000 prevented (100% substitution) in Portugal. In Denmark, the maximum health gain would also be observed with 100% substitution of beef by pulses, with an estimated 39.7 DALYs per 100,000 prevented; 25% substitution would lead to 10.7 years of life saved per 100,000. The reduction of the risk of CRC was the largest contribution for the positive health effect in Denmark in all scenarios, whereas prevention of IHD had the largest impact in Portugal. The sustainability assessment showed positive economic impacts of the substitutions in the two countries. Beef production seems more resilient and stable than pulses in the two countries. The measured environmental impacts of the beef value chain were consistently higher in the two countries. Beef production was estimated to have a higher contribution to environmental degradation and resource use, with GHG emissions, eutrophying emissions, and land use showing the greatest differences between the two food groups.

Through two rounds of consultations, experts from fields of risk assessment, management and communication shared their experience and perspectives for the development of a novel framework, which is expected to be adopted and applied in future evaluations. A questionnaire to assess individual capacities within an Organization’s staff and one to be filled at the level of Organization were developed in the context of ALTERNATIVA project.

The qualitative research approach applied to prioritize concepts and aspects that need to be addressed through an exploratory questionnaire assessing capacities at either organisational or individual level revealed the stakeholders' perceptions and attitudes. Concepts (e.g. risk and benefit) and methods (e.g. the step-wise RBA approach) need to be well defined, transparent and standardized. It is worth noting though that stakeholders participating in this exercise were reluctant to the integration of sustainability aspects related to the environment, social and economic dimensions into current RBA frameworks which primarily rely on addressing health risks and benefits.

Altogether, the results and outputs obtained in the context of ALTERNATIVA project constitute, undoubtably, crucial, and urgent contributions that could better support future decisions in the context of alternative protein sources, and consequently affecting the health of consumers and the sustainability of food systems. Thus, this project is a key example of the importance of multidisciplinary approaches in tackling the current and future societal challenges, in this particular case, the necessary and urgent food systems transition.

## 4.1 Impact assessment

The project ALTERNATIVA had contributed significantly four dimensions: i) capacity-building, in the context of risk-benefit assessment, sustainability assessment, and the application of MCDA; ii) methodologically, through the development of the protocol of integration of risk-benefit and sustainability assessment; iii) obtaining robust scientific evidence about the impact of substituting the consumption of red meat by pulses; and, iv) contributing to the current understanding and attitudes towards the use of risk-benefit assessment and sustainability assessment of organizations dealing with these issues.

Building on the methodologies developed under the previous EFSA-supported projects, RiskBenefit4EU and NovRBA, ALTERNATIVA advanced the previous approaches, integrating in the methodology different dimensions of sustainability, i.e., economic, social and environment. In this context, the capacity building activities, namely the training activities but also the application of knowledge to a practical case study, improved the knowledge of the consortium members in the main areas of the project.

The protocol developed, intended to establish a framework that could support the integration of health risk-benefit assessment of foods and sustainability assessment, constitutes also an important contribution, that could impact significantly the obtainment of scientific evidence regarding, primarily, the alternative protein sources, but, most broadly, supporting changes in the food systems, aligned with better decisions for health and sustainability.

In the context of selecting alternative protein sources to animal-based products, the performed case study also represents an important achievement. Pulses are indicated, in the context of western consumption habits, as a natural alternative to animal-based food products. The performed case study evidenced the impact of a potential substitution of the consumption of red meat by pulses (a mix of different pulses considered for both countries). The obtained quantitative results of the health, environmental and socio-economic revealed impacts of four possible scenarios of substitution of consumption of beef by pulses in the

Portuguese and the Danish populations. They all point to the benefits of doing such transitions, which supports the further development of policies for promoting a decrease in consumption of beef (and other meat products) and increase in intake of plant-based sources of protein.

Regarding sustainability, Figure 13 and Figure 14 provide the graphical representation of the sustainability assessment, respectively, in Denmark and Portugal, of substituting red meat with pulses. Each branch presents the performance of pulses compared to beef (100 in the chart) for each sustainability indicator. A value of 151 for indicator "Net result", indicates that for such indicator the performance of pulses is 51% higher than for beef. For the environmental and affordability indicators, for which lower is better, the opposite of the difference is displayed (e.g., +20% when the carbon footprint is 20% lower) (Arfini & Bellassen, 2019). As could be seen in the figures, the scenarios presenting higher consumption of pulses, compared to red meat, are expected to provide more benefits in terms of the sustainability indicators considered.

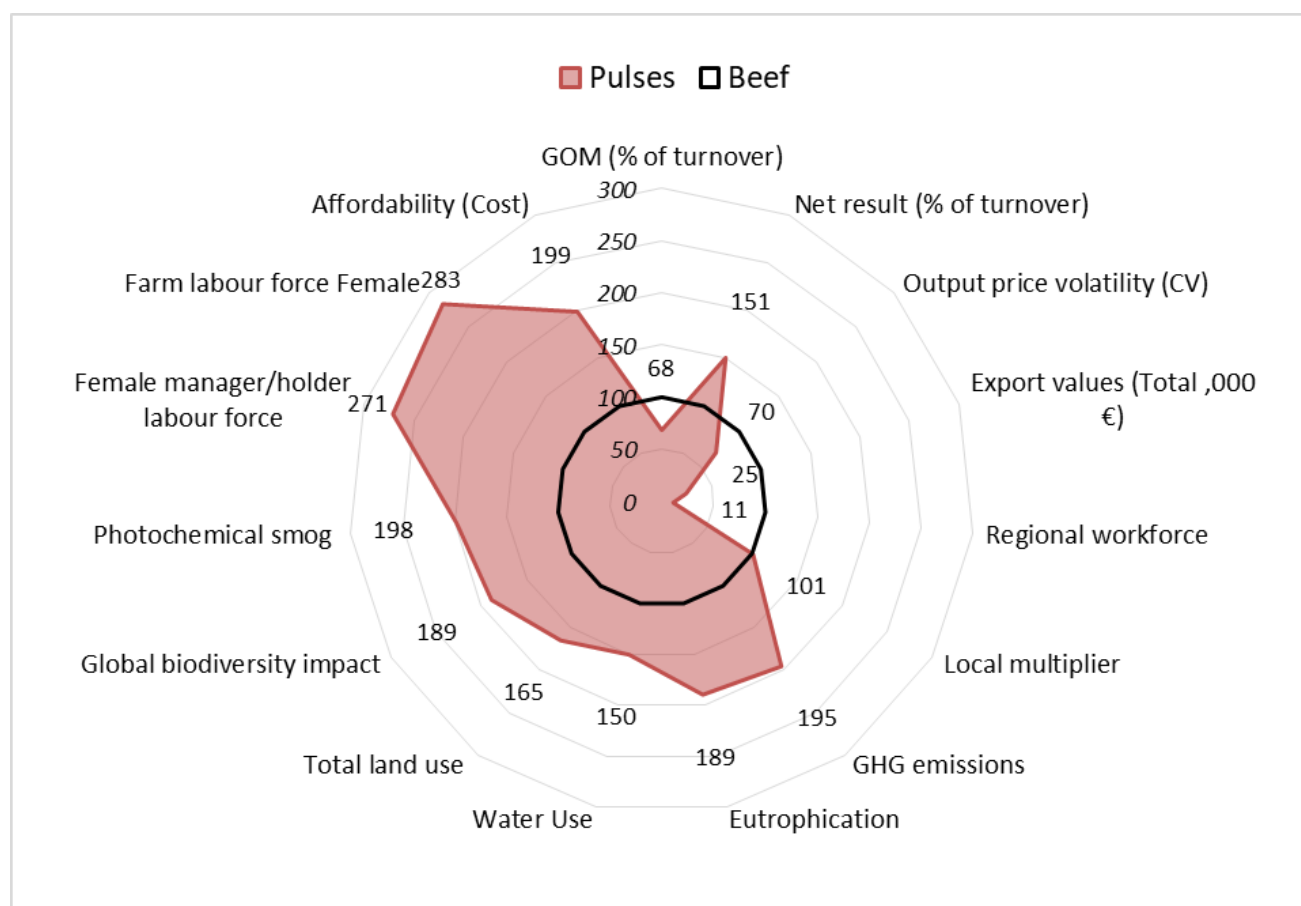


Figure 13. Sustainability impact assessment of pulses compared to beef in Denmark (value 100 in the chart = beef performance).

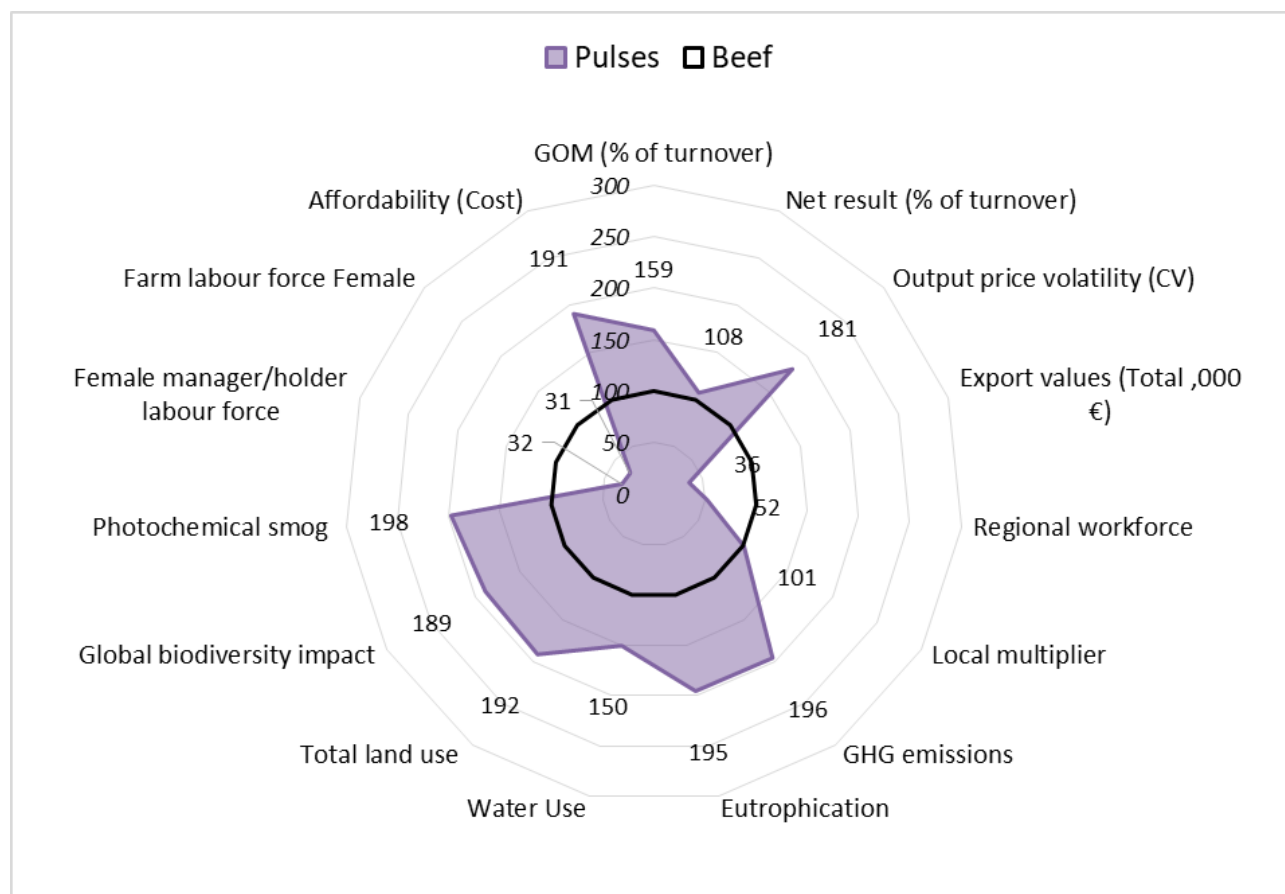


Figure 14. Sustainability Impact assessment of pulses compared to beef in Portugal (value 100 in the chart = beef performance).

Through a collaborative process involving food scientists, researchers, public officials in the food sector and a psychologist with expertise in conducting inductive research, the ALTERNATIVA project applied methods of exploratory research (i.e. focus groups) to identify items within each RBA domain that are relevant for a questionnaire to characterize the current understanding and attitudes towards the use of risk-benefit assessment and sustainability assessment of organizations dealing with these issues. Exploratory research revealed limitations in the knowledge of RBA principles and methods. The broad perception participants described was primarily relying on their common understanding of the four pillars of risk assessment. Furthermore, they all stressed the importance of training towards harmonized approaches, clear outcomes and guidance on how to communicate the message to consumers. The questionnaire will permit to understand how the different stakeholders will be positioned regarding the use of these tools (RBA and sustainability assessment) to address the current and future problems, gather evidence, and take decisions.

Overall, ALTERNATIVA project has advanced the already established methodologies in the context of the impact assessment of foods, in terms of health and sustainability. But this project also created evidence regarding the impact of doing important changes in the consumption patterns of specific foods, traditionally indicated to have a higher impact on

health and sustainability (e.g. animal-based products as red meat). Altogether, the obtained outputs have the potential to open new research avenues that will benefit humans, animals and the environment.

## 5 Recommendations

Through the ALTERNATIVA project, the consortium identified a set of opportunities and limitations that were the basis for the establishment of these recommendations:

- Improvements in the data quality and availability: the integration of health risk-benefit assessment of foods and sustainability assessment requires a significant amount of data (e.g. compositional data, dose-response, data to feed the sustainability indicators), that are not always available and/or presenting the proper quality. This fact implies assumptions or the use of proxy that increases the associated uncertainties;
- Improvement of knowledge on health (RBA) and sustainability impact assessment: given the multidisciplinary nature of the sustainability and health issues, as well as the non-linear nature of the problems associated, we could also recommend improving the competencies and skills, in particular in the national and European agencies, regarding the sustainability and risk-benefit assessment procedures, as well as the competences for MCDA methodologies;
- Dissemination of project outputs and main results: as referred before, ALTERNATIVA project produced significant and important outputs and results that certainly could benefit decisions related to alternative protein sources and food system transition;
- Increased use of multi-criteria decision analysis methodology: to support decisions, similarly to what was developed in the context of ALTERNATIVA project, the use of MCDA should be democratized;
- Implementation of the questionnaire developed in different Member States: this questionnaire will serve the goal of characterizing the current understanding of the different stakeholders regarding the use of RBA and sustainability assessment to address the current and future problems, gathering evidence, and taking decisions.

Overall, these recommendations resulted from the experience of the consortium dealing with the challenges faced during the process. They should be seen as priorities for generating enhanced pieces of evidence that should further support future developments of the integration of health risk-benefit assessment of foods and sustainability assessment.

## References

- Afshin, A., Micha, R., Khatibzadeh, S., & Mozaffarian, D. (2014). Consumption of nuts and legumes and risk of incident ischemic heart disease, stroke, and diabetes: a systematic review and meta-analysis. *The American Journal of Clinical Nutrition*, 100(1), 278–288. <https://doi.org/10.3945/ajcn.113.076901>
- Aiking, H., & de Boer, J. (2018). The next protein transition. *Trends in Food Science & Technology*, July, 0–1. <https://doi.org/10.1016/j.tifs.2018.07.008>
- Aldaya, M. M., Ibañez, F. C., Domínguez-Lacueva, P., Murillo-Arbizu, M. T., Rubio-Varas, M., Soret, B., & Beriain, M. J. (2021). Indicators and Recommendations for Assessing Sustainable Healthy Diets. *Foods*, 10(5), 999. <https://doi.org/10.3390/foods10050999>
- Ambikapathi, R., Schneider, K. R., Davis, B., Herrero, M., Winters, P., & Fanzo, J. C. (2022). Global food systems transitions have enabled affordable diets but had less favourable outcomes for nutrition, environmental health, inclusion and equity. *Nature Food*, 3(9), 764–779. <https://doi.org/10.1038/s43016-022-00588-7>
- Arfini, F., & Bellassen, V. (2019). *Sustainability of European Food Quality Schemes. Multi-Performance, Structure, and Governance of PDO, PGI, and Organic Agri-Food Systems*. Springer Nature Switzerland. <https://www.springer.com/gp/book/9783030275075>
- Assunção, R., Alvito, P., Brazão, R., Carmona, P., Fernandes, P., Jakobsen, L. S., Lopes, C., Martins, C., Membré, J.-M., Monteiro, S., Nabais, P., Thomsen, S. T., Torres, D., Viegas, S., Pires, S. M., & Boué, G. (2019). Building capacity in risk-benefit assessment of foods: Lessons learned from the RB4EU project. *Trends in Food Science & Technology*, 91, 541–548. <https://doi.org/10.1016/j.tifs.2019.07.028>
- Boué, G., Guillou, S., Antignac, J.-P., Bizec, B., & Membré, J.-M. (2015). Public Health Risk-benefit Assessment Associated with Food Consumption—A Review. *European Journal of Nutrition & Food Safety*, 5(1), 32–58. <https://doi.org/10.9734/EJNFS/2015/12285>
- Boué, G., Ververis, E., Niforou, A., Federighi, M., Pires, S. M., Poulsen, M., Thomsen, S. T., & Naska, A. (2022). Risk–Benefit assessment of foods: Development of a methodological framework for the harmonized selection of nutritional, microbiological, and toxicological components. *Frontiers in Nutrition*, 9. <https://doi.org/10.3389/fnut.2022.951369>
- Clark, M. A., Springmann, M., Hill, J., & Tilman, D. (2019). Multiple health and environmental impacts of foods. *Proceedings of the National Academy of Sciences*, 116(46), 23357–23362. <https://doi.org/10.1073/pnas.1906908116>
- Cucurachi, S., Scherer, L., Guinée, J., & Tukker, A. (2019). Life Cycle Assessment of Food Systems. *One Earth*, 1(3), 292–297. <https://doi.org/10.1016/j.oneear.2019.10.014>
- Dentoni, D., Cucchi, C., Roglic, M., Lubberink, R., Bender-Salazar, R., & Manyise, T. (2023). Systems Thinking, Mapping and Change in Food and Agriculture. *Bio-Based and Applied Economics*, 11(4), 277–301. <https://doi.org/10.36253/bae-13930>
- Donati, M., Wilkinson, A., Veneziani, M., Antonioli, F., Arfini, F., Bodini, A., Amilien, V., Csillag, P., Ferrer-Pérez, H., Gkatsikos, A., Gauvrit, L., Gil, C., Hoàng, V., Knutsen

Steinnes, K., Lilavanichakul, A., Mattas, K., Napisintuwong, O., Nguyễn, A., Nguyen, M., ... Bellassen, V. (2021). Economic Spill-Over of Food Quality Schemes on Their Territory. *Journal of Agricultural & Food Industrial Organization*, 19(2), 95–111. <https://doi.org/10.1515/jafio-2019-0046>

EFSA Scientific Committee. (2010). Guidance on human health risk-benefit assessment of foods. *EFSA Journal*, 8(7), 1673. <https://doi.org/10.2093/j.efsa.2010.1673>

European Commission. (2019). *The European Green Deal*.

Fabricius, F. A., Thomsen, S. T., Fagt, S., & Nauta, M. (2021). The health impact of substituting unprocessed red meat by pulses in the Danish diet. *European Journal of Nutrition*, 60(6), 3107–3118. <https://doi.org/10.1007/s00394-021-02495-2>

FAO. (2013a). *SAFA indicators*. Food and Agriculture Organization of the United Nations (FAO). [http://www.fao.org/fileadmin/templates/nr/sustainability\\_pathways/docs/SAFA\\_Indicators\\_final\\_19122013.pdf](http://www.fao.org/fileadmin/templates/nr/sustainability_pathways/docs/SAFA_Indicators_final_19122013.pdf)

FAO. (2013b). *SAFA Sustainability Assessment of Food and Agriculture Systems: Guidelines, version 3.0*. <https://www.fao.org/nr/sustainability/sustainability-assessments-safa/en/>

Fasolin, L. H., Pereira, R. N., Pinheiro, A. C., Martins, J. T., Andrade, C. C. P., Ramos, O. L., & Vicente, A. A. (2019). Emergent food proteins – Towards sustainability, health and innovation. *Food Research International*, 125(July), 108586. <https://doi.org/10.1016/j.foodres.2019.108586>

Gonzalez Fischer, C., & Garnett, T. (2016). *Plates, Pyramids, Planet: Developments in National Healthy and Sustainable Dietary Guidelines: A State of Play Assessment*.

Henchion, M., Hayes, M., Mullen, A., Fenelon, M., & Tiwari, B. (2017). Future Protein Supply and Demand: Strategies and Factors Influencing a Sustainable Equilibrium. *Foods*, 6(7), 53. <https://doi.org/10.3390/foods6070053>

Hoekstra, J., Hart, A., Owen, H., Zeilmaker, M., Bokkers, B., Thorgilsson, B., & Gunnlaugsdottir, H. (2013). Fish, contaminants and human health: Quantifying and weighing benefits and risks. *Food and Chemical Toxicology*, 54, 18–29. <https://doi.org/10.1016/j.fct.2012.01.013>

Hollander, A., De Jonge, R., Biesbroek, S., Hoekstra, J., & Zijp, M. C. (2019). Exploring solutions for healthy, safe, and sustainable fatty acids (EPA and DHA) consumption in The Netherlands. *Sustainability Science*, 14(2), 303–313. <https://doi.org/10.1007/s11625-018-0607-9>

Irz, X., Jensen, J. D., Leroy, P., Réquillart, V., & Soler, L.-G. (2019). Promoting climate-friendly diets: What should we tell consumers in Denmark, Finland and France? *Environmental Science & Policy*, 99, 169–177. <https://doi.org/10.1016/j.envsci.2019.05.006>

Joint Research Centre-European Commission. (2008). *Handbook on constructing composite indicators: methodology and user guide*. OECD publishing.

- Lonnie, M., Hooker, E., Brunstrom, J., Corfe, B., Green, M., Watson, A., Williams, E., Stevenson, E., Penson, S., & Johnstone, A. (2018). Protein for Life: Review of Optimal Protein Intake, Sustainable Dietary Sources and the Effect on Appetite in Ageing Adults. *Nutrients*, 10(3), 360. <https://doi.org/10.3390/nu10030360>
- Lopes, C., Torres, D., Oliveira, A., Severo, M., Guimaraes, S., Alarcão, V., Ramos, E., Rodrigues, S., Vilela, S., Oliveira, L., Mota, J., Teixeira, P. J., Nicola, P. J., Soares, S., & Andersen, L. F. (2018). National Food, Nutrition, and Physical Activity Survey of the Portuguese General Population (2015-2016): Protocol for Design and Development. *JMIR Research Protocols*, 7(2), e42. <https://doi.org/10.2196/resprot.8990>
- Lynch, H., Johnston, C., & Wharton, C. (2018). Plant-Based Diets: Considerations for Environmental Impact, Protein Quality, and Exercise Performance. *Nutrients*, 10(12), 1841. <https://doi.org/10.3390/nu10121841>
- Malapit, H., Pinkstaff, C., Sproule, K., & et al. (2017). The Abbreviated Women's Empowerment in Agriculture Index (A-WEAI). *IFPRI Discussion Paper 1647*.
- Mancini, S., Sogari, G., Menozzi, D., Nuvoloni, R., Torracca, B., Moruzzo, R., & Paci, G. (2019). Factors Predicting the Intention of Eating an Insect-Based Product. *Foods*, 8(7), 270. <https://doi.org/10.3390/foods8070270>
- Mason-D'Croz, D., Barnhill, A., Bernstein, J., Bogard, J., Dennis, G., Dixon, P., Fanzo, J., Herrero, M., McLaren, R., Palmer, J., Rieder, T., Rimmer, M., & Faden, R. (2022). Ethical and economic implications of the adoption of novel plant-based beef substitutes in the USA: a general equilibrium modelling study. *The Lancet Planetary Health*, 6(8), e658–e669. [https://doi.org/10.1016/S2542-5196\(22\)00169-3](https://doi.org/10.1016/S2542-5196(22)00169-3)
- Membré, J. M., Santillana Farakos, S., & Nauta, M. (2021). Risk-benefit analysis in food safety and nutrition. *Current Opinion in Food Science*, 39, 76–82. <https://doi.org/10.1016/j.cofs.2020.12.009>
- Menozzi, D., Sogari, G., Veneziani, M., Simoni, E., & Mora, C. (2017). Eating novel foods: An application of the Theory of Planned Behaviour to predict the consumption of an insect-based product. *Food Quality and Preference*, 59, 27–34. <https://doi.org/10.1016/j.foodqual.2017.02.001>
- Nadathur, S. R., Wanasundara, J. P. D., & Scanlin, L. (2017). Proteins in the Diet. In *Sustainable Protein Sources* (pp. 1–19). Elsevier. <https://doi.org/10.1016/B978-0-12-802778-3.00001-9>
- Naska, A., Ververis, E., Niforou, A., Pires, S. M., Poulsen, M., Jakobsen, L. S., Becker, N., Lohmann, M., Tesson, V., Federighi, M., & Boué, G. (2022). Novel foods as red meat replacers – an insight using Risk Benefit Assessment methods (the NovRBA project). *EFSA Supporting Publications*, 19(5). <https://doi.org/10.2903/SP.EFSA.2022.EN-7316>
- Norat, T., Vieira, A. R., Abar, L., Aune, D., Polemiti, E., Chan, D., & Vingeliene, S. (2017). *World Cancer Research Fund international systematic literature review: The associations between food, nutrition and physical activity and the risk of colorectal cancer*.

- Nordic Council of Ministers. (2014). Nordic Nutrition Recommendations 2012: Integrating nutrition and physical activity. In *Nordic Nutrition Recommendations 2012* (Vol. 5, Issue 11). <https://doi.org/10.6027/Nord2014-002>
- Pan, A., Sun, Q., Bernstein, A. M., Schulze, M. B., Manson, J. E., Willett, W. C., & Hu, F. B. (2011). Red meat consumption and risk of type 2 diabetes: 3 cohorts of US adults and an updated meta-analysis. *The American Journal of Clinical Nutrition*, 94(4), 1088–1096. <https://doi.org/10.3945/ajcn.111.018978>
- Pedersen, A. N., Christensen, T., Matthiessen, J., Knudsen, V. K., Rosenlund-Sørensen, M., Biloft-Jensen, A., Hinsch, H. J., Ygil, K. H., Kørup, K., Saxholt, E., Trolle, E., Søndergaard, A. B., & Fagt, S. (2015). Dietary habits in Denmark 2011-2013. Main results. In *Danskernes kostvaner 2011-2013* (p. 210).
- Pires, S. M., Boué, G., Boobis, A., Eneroth, H., Hoekstra, J., Membré, J.-M., Persson, I. M., Poulsen, M., Ruzante, J., van Klaveren, J., Thomsen, S. T., & Nauta, M. J. (2019). Risk Benefit Assessment of foods: Key findings from an international workshop. *Food Research International*, 116, 859–869. <https://doi.org/10.1016/j.foodres.2018.09.021>
- Rogge, N. (2018). Composite indicators as generalized benefit-of-the-doubt weighted averages. *European Journal of Operational Research*, 267(1), 381–392. <https://doi.org/10.1016/j.ejor.2017.11.048>
- Sala, S., Anton, A., McLaren, S. J., Notarnicola, B., Saouter, E., & Sonesson, U. (2017). In quest of reducing the environmental impacts of food production and consumption. *Journal of Cleaner Production*, 140, 387–398. <https://doi.org/10.1016/j.jclepro.2016.09.054>
- Smetana, S. M., Bornkessel, S., & Heinz, V. (2019). A Path From Sustainable Nutrition to Nutritional Sustainability of Complex Food Systems. *Frontiers in Nutrition*, 6(April), 1–6. <https://doi.org/10.3389/fnut.2019.00039>
- Sogari, G., Menozzi, D., & Mora, C. (2019). The food neophobia scale and young adults' intention to eat insect products. *International Journal of Consumer Studies*, 43(1), 68–76. <https://doi.org/10.1111/ijcs.12485>
- Spector, T. D., & Gardner, C. D. (2020). Challenges and opportunities for better nutrition science—an essay by Tim Spector and Christopher Gardner. *BMJ*, 369, m2470. <https://doi.org/10.1136/bmj.m2470>
- Springmann, M., Wiebe, K., Mason-D'Croz, D., Sulser, T. B., Rayner, M., & Scarborough, P. (2018). Health and nutritional aspects of sustainable diet strategies and their association with environmental impacts: a global modelling analysis with country-level detail. *The Lancet Planetary Health*, 2(10), e451–e461. [https://doi.org/10.1016/S2542-5196\(18\)30206-7](https://doi.org/10.1016/S2542-5196(18)30206-7)
- Takacs, B., & Borrion, A. (2020). The Use of Life Cycle-Based Approaches in the Food Service Sector to Improve Sustainability: A Systematic Review. *Sustainability*, 12(9), 3504. <https://doi.org/10.3390/su12093504>

Tian, J. J., Bryksa, B. C., & Yada, R. Y. (2016). Feeding the world into the future – food and nutrition security: the role of food science and technology †. *Frontiers in Life Science*, 9(3), 155–166. <https://doi.org/10.1080/21553769.2016.1174958>

United Nations. (2015). *Transforming our World: The 2030 Agenda for Sustainable Development*.

van Asselt, E. D., van Bussel, L. G. J., van der Voet, H., van der Heijden, G. W. A. M., Tromp, S. O., Rijgersberg, H., van Evert, F., Van Wagenberg, C. P. A., & van der Fels-Klerx, H. J. (2014). A protocol for evaluating the sustainability of agri-food production systems—A case study on potato production in peri-urban agriculture in The Netherlands. *Ecological Indicators*, 43, 315–321. <https://doi.org/10.1016/j.ecolind.2014.02.027>

van der Goot, A. J., Pelgrom, P. J. M., Berghout, J. A. M., Geerts, M. E. J., Jankowiak, L., Hardt, N. A., Keijer, J., Schutyser, M. A. I., Nikiforidis, C. V., & Boom, R. M. (2016). Concepts for further sustainable production of foods. *Journal of Food Engineering*, 168, 42–51. <https://doi.org/10.1016/j.jfoodeng.2015.07.010>

van der Spiegel, M., Noordam, M. Y., & van der Fels-Klerx, H. J. (2013). Safety of Novel Protein Sources (Insects, Microalgae, Seaweed, Duckweed, and Rapeseed) and Legislative Aspects for Their Application in Food and Feed Production. *Comprehensive Reviews in Food Science and Food Safety*, 12(6), 662–678. <https://doi.org/10.1111/1541-4337.12032>

van Kleef, E., Frewer, L. J., Chrysoschoidis, G. M., Houghton, J. R., Korzen-Bohr, S., Krystallis, T., Lassen, J., Pfenning, U., & Rowe, G. (2006). Perceptions of food risk management among key stakeholders: Results from a cross-European study. *Appetite*, 47(1), 46–63. <https://doi.org/10.1016/j.appet.2006.02.002>

Willett, W., Rockström, J., Loken, B., Springmann, M., Lang, T., Vermeulen, S., Garnett, T., Tilman, D., DeClerck, F., Wood, A., Jonell, M., Clark, M., Gordon, L. J., Fanzo, J., Hawkes, C., Zurayk, R., Rivera, J. A., De Vries, W., Majele Sibanda, L., ... Murray, C. J. L. (2019). Food in the Anthropocene: the EAT–Lancet Commission on healthy diets from sustainable food systems. *The Lancet*, 393(10170), 447–492. [https://doi.org/10.1016/S0140-6736\(18\)31788-4](https://doi.org/10.1016/S0140-6736(18)31788-4)

Zhou, L., Tokos, H., Krajnc, D., & Yang, Y. (2012). Sustainability performance evaluation in industry by composite sustainability index. *Clean Technologies and Environmental Policy*, 14(5), 789–803. <https://doi.org/10.1007/s10098-012-0454-9>

## Abbreviations

|          |                                                                                    |
|----------|------------------------------------------------------------------------------------|
| ANSVSA   | National Sanitary Veterinary and Food Safety Authority                             |
| CRC      | Colorectal cancer                                                                  |
| DALY     | Disability-adjusted life year                                                      |
| DANSDA   | Danish National Survey of Diet and Physical Activity                               |
| DK       | Denmark                                                                            |
| DTU      | Technical University of Denmark                                                    |
| DTU Food | National Food Institute, Technical University of Denmark                           |
| EFSA     | European Food Safety Authority                                                     |
| EU       | European Union                                                                     |
| FAO      | Food and Agriculture Organisation of the United Nations                            |
| GHG      | Greenhouse gas                                                                     |
| GOM      | Gross Operating Margin                                                             |
| IAN-AF   | National Food, Nutrition and Physical Activity Survey of the Portuguese Population |
| IHD      | Ischaemic heart disease                                                            |
| INRAE    | French National Institute for Agriculture, Food and Environment                    |
| INSA     | Instituto Nacional de Saúde Doutor Ricardo Jorge                                   |
| LCA      | Life cycle assessment                                                              |
| MCDA     | Multi-Criteria Decision Analysis                                                   |
| NKUA     | National and Kapodistrian University of Athens                                     |
| PPP      | Purchasing Power Parity                                                            |
| PT       | Portugal                                                                           |
| RBA      | Health risk-benefit assessment                                                     |
| SDGs     | Sustainable Development Goals                                                      |
| UN       | United Nations                                                                     |
| UNIPR    | University of Parma                                                                |
| WHO      | World Health Organization                                                          |
| WP       | Work packages                                                                      |
| YLD      | Years of life lived with disability                                                |
| YLL      | Years of life lost                                                                 |

## ANNEXES

### Annex 1 – Action plan (Deliverable 1)

Assunção, R. (2024). ALTERNATIVA Project: Action Plan. Zenodo.  
<https://doi.org/10.5281/zenodo.13799697>

### Annex 2 – Training plan (Deliverable 2)

Assunção, R. (2024). ALTERNATIVA Project: Training Plan. Zenodo.  
<https://doi.org/10.5281/zenodo.13799748>

### Annex 3 – Policy brief (Deliverable 7)

Assunção, R. (2024). ALTERNATIVA Project: Policy Brief. Zenodo.  
<https://doi.org/10.5281/zenodo.13799750>

### Annex 4 – National publication 1

For the publication: <http://hdl.handle.net/10400.18/8632>

### Annex 5 – National publication 2

For the publication: <http://hdl.handle.net/10400.18/8882>

### Annex 6 – ALTERNATIVA workshop programme

Assunção, R. (2024). ALTERNATIVA WORKSHOP – FOOD AND SUSTAINABILITY. Zenodo.  
<https://doi.org/10.5281/zenodo.13799762>